

SC ENDSEM

05/16/2026 19:39:26

Generate a **SPPU University endsem exam-ready answer for Social...

Use this master prompt for generating high-scoring answers in SPPU BE IT...

📘 UNIT 3 → Q1 & Q2 What is data mining? Draw and explain architecture of ...

Here is the SPPU exam-ready answer for Social Computing (SC)....

What is classification? Explain with diagram / Explain any one classification...

:::writing{variant="document" id="68425"} 📘 What is Classification? Explain...

Explain Hierarchical clustering algorithm with single linkage clustering.

:::writing{variant="document" id="41736"} 📘 Explain Hierarchical Clustering...

What is clustering? Explain any clustering algorithm with example.

:::writing{variant="document" id="29641"} 📘 What is Clustering? Explain Any...

Explain data mining methods for social media. 8

:::writing{variant="document" id="57328"} 📘 Explain Data Mining Methods f...

Write note on Text mining in social networks. 8

:::writing{variant="document" id="61842"} 📘 Write Note on Text Mining in...

Explain data representation methods for social media. 8 Nov 2024 ★ What is...

Here are SPPU exam-ready answers for all four questions....

📘 UNIT 4 → Q3 & Q4 What is Homophily? Explain how to measure Homophil...

:::writing{variant="document" id="74215"} 📘 What is Homophily? Explain Ho...

What is Influence? Explain how to measure influence in social computing. 8/9

:::writing{variant="document" id="58314"} 📘 What is Influence? Explain How...

Explain Shuffle Test and Edge Reversal Test. 9

:::writing{variant="document" id="46182"} 📘 Explain Shuffle Test and Edge...

Explain Randomization Test with example / in detail. 8/9

:::writing{variant="document" id="81547"} 📘 Explain Randomization Test wit...

What is Assortativity? Explain any one technique to measure assortativity. 9

:::writing{variant="document" id="39264"} 📘 What is Assortativity? Explain A...

Distinguish Influence and Homophily. 9 Apr 2023 ★ What is a shuffle test...

:::writing{variant="document" id="52847"} 📖 Distinguish Between Influence...

📖 UNIT 5 → Q5 & Q6 ⭐⭐⭐ What is Individual Behavior? Explain individual...

:::writing{variant="document" id="68492"} 📖 What is Individual Behavior?...

What is Collective Behavior Analysis? Explain User Migration in social media. 9

:::writing{variant="document" id="73158"} 📖 What is Collective Behavior...

Write short note on Scraping, Parsing and Crawling the web. 9

:::writing{variant="document" id="84267"} 📖 Write Short Note on Scraping,...

Describe breadth-first search in web crawling and pros & cons. 9

:::writing{variant="document" id="61395"} 📖 Describe Breadth-First Search i...

Explain query human language data with TF-IDF. 9

:::writing{variant="document" id="45721"} 📖 Explain Query Human Languag...



Generate a **SPPU University endsem exam-ready answer** for Social Computing (SC) that helps a BE IT student score maximum marks in theory exams.

Main Goal

The answer should help a student:

- * easily understand the concept,
- * remember it quickly,
- * reproduce it in SPPU university exams,
- * and write a high-scoring answer even with weak theory knowledge.

The answer should feel like:

👉 "Now I can easily write this answer in the SPPU exam."

Very Important Teaching Style

- * Write in **very simple English**.
- * Teach like explaining to a beginner or a 10-year-old student.
- * Avoid difficult technical jargon.
- * Use **short sentences** and **simple words**.
- * Explain concepts step-by-step logically.
- * Use **point-wise format** for easy exam writing.
- * Make the answer easy for last-minute revision.

Real-Life Examples (VERY IMPORTANT)

For every concept, include **simple real-life examples** from:

- * Instagram
- * WhatsApp
- * Facebook
- * Twitter/X
- * YouTube
- * LinkedIn

- * Online communities
- * Gaming communities
- * Reddit
- * Telegram
- * E-learning platforms

After giving the example:

- * clearly explain HOW the example relates to the concept.

Exam-Oriented Requirements

The answer should include:

1 Simple Introduction (2–3 lines)

Explain the concept in very easy language.

2 Main Concept Explanation

Explain:

- * meaning,
 - * working,
 - * process,
 - * components,
 - * features,
 - * advantages/disadvantages,
 - * applications,
- step-by-step in simple points.

3 Real-Life Social Media Example

Give practical examples from social media platforms.

Example:

- * Instagram recommendations
- * WhatsApp groups
- * Twitter trends
- * YouTube comments
- * Facebook communities

4 Explain the Example Clearly

Explain how the example connects to the concept.

5 Simple Diagram (VERY IMPORTANT)

Draw easy exam-style diagrams using text.

Example:

```text

User → Social Platform → Data Sharing → Interaction → Community

```

After the diagram:

- * explain every part simply.

6 Key Points / Features

Write important exam points in bullets.

7 Advantages and Disadvantages (if applicable)

Write in table or point format.

8 Applications / Uses

Mention practical uses in real life.

9 Memory Tricks / Mnemonics (VERY IMPORTANT)

Give easy tricks to remember difficult concepts quickly.

Example:

- * short forms
 - * funny sentence tricks
 - * keyword tricks
-

10 Exam Writing Tips

Mention:

- * what to underline,
 - * which keywords are important,
 - * which diagrams to draw,
 - * which examples give extra marks.
-

11 Short Conclusion (2 lines)

Give a clean final summary.

12 Last Minute Revision Points

Add quick revision bullets at the end.

1 3 Viva Questions with Answers

Add 3–5 small viva questions with very short answers.

Presentation Style

- * Use headings and subheadings clearly.
- * Use bullet points.
- * Use spacing properly.
- * Make answer look neat like topper notes.

Important Extra Instructions

- ✓ Include keywords students should underline in exam.
- ✓ Use SPPU theory-paper style answer writing.
- ✓ Add comparison tables wherever useful.
- ✓ Add diagrams whenever possible.
- ✓ Keep answers easy to memorize after one reading.
- ✓ Prefer exam-friendly language over textbook language.

Subject:

Social Computing (SC)

Question:

[Paste your SC question here]

Marks:

[5 / 8 / 10]

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Use this master prompt for generating high-scoring answers in **SPPU BE IT Social Computing (SC)** theory exams.

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SPPU Social Computing (SC) Endsem Answer Generator Prompt

Generate a **SPPU University endsem exam-ready answer for Social Computing (SC)** that helps a BE IT student score maximum marks in theory exams.



Main Goal

The answer should help a student:

- easily understand the concept,
- remember it quickly,
- reproduce it in SPPU university exams,
- and write a high-scoring answer even with weak theory knowledge.

The answer should feel like:

👉 "Now I can easily write this answer in the SPPU exam."



Very Important Teaching Style

- Write in **very simple English**.
- Teach like explaining to a beginner or a 10-year-old student.
- Avoid difficult technical jargon.

- Use short sentences and simple words.
 - Explain concepts step-by-step logically.
 - Use point-wise format for easy exam writing.
 - Make the answer easy for last-minute revision.
-

Real-Life Examples (VERY IMPORTANT)

For every concept, include simple real-life examples from:

- Instagram
- WhatsApp
- Facebook
- Twitter/X
- YouTube
- LinkedIn
- Reddit
- Telegram
- Online communities
- Gaming communities
- E-learning platforms

After giving the example:

- clearly explain HOW the example relates to the concept.
-



Exam-Oriented Requirements

The answer MUST contain the following sections:

1 Simple Introduction (2–3 lines)

Explain the concept in very easy language.

2 Main Concept Explanation

Explain in simple points:

- meaning
- definition
- working
- process
- components
- characteristics
- features
- advantages
- disadvantages
- applications

Explain step-by-step logically.

3 Real-Life Social Media Example

Give practical examples from social platforms.

Examples:

- Instagram recommendation system
 - WhatsApp groups
 - Facebook communities
 - Twitter/X trending hashtags
 - YouTube comments and likes
 - LinkedIn networking
 - Reddit discussion communities
 - Telegram channels
-

4 Explain the Example Clearly

Clearly explain:

- how the example works,
 - how users interact,
 - and how it relates to the concept.
-

5 Simple Diagram (VERY IMPORTANT)

Draw easy exam-style diagrams using text.

Example:

User → Social Platform → Data Sharing → Interaction → Community

OR

User → Post → Likes/Comments → Sharing → Viral Content

After the diagram:

- explain every block in simple words.

6 Key Points / Features

Write important exam points in bullets.

Use short and memorable points.

7 Advantages and Disadvantages

Write in table format whenever possible.

Example:

Advantages	Disadvantages
Fast communication	Privacy issues
Easy collaboration	Fake information

8 Applications / Uses

Mention practical real-life uses such as:

- Online learning
- Social networking
- Marketing
- E-commerce
- Team collaboration
- Community building
- Online gaming
- Crowdsourcing
- Public awareness

9 Memory Tricks / Mnemonics (VERY IMPORTANT)

Give easy tricks to remember concepts quickly.

Examples:

- Short forms
- Funny keyword tricks
- First-letter mnemonics

Example:

For SC features:

👉 "CICS"

- C → Communication
 - I → Interaction
 - C → Collaboration
 - S → Sharing
-

10 Exam Writing Tips

Mention:

- what keywords to underline,
 - which headings are important,
 - which diagram to draw,
 - which real-life example gives extra marks,
 - how to write the answer neatly.
-

1 1 Short Conclusion (2 lines)

Give a simple and clean final summary.

1 2 Last Minute Revision Points

Add 5–10 quick revision bullets for fast revision before exam.

13 Viva Questions with Answers

Add 3–5 short viva questions with very small answers.

Example:

Q1. What is Social Computing?

Ans: Use of social interaction with computer systems.

Q2. Give one example of SC.

Ans: Facebook.

Presentation Style

- Use proper headings and subheadings.
- Use bullet points.
- Keep spacing properly.
- Make the answer look like topper notes.
- Prefer point-wise writing over paragraphs.

Important Extra Instructions

- ✓ Include keywords students should underline in exam.
- ✓ Use SPPU theory-paper writing style.
- ✓ Add comparison tables wherever useful.

- ✓ Add simple diagrams whenever possible.
 - ✓ Keep answers easy to memorize after one reading.
 - ✓ Prefer exam-friendly language over textbook language.
 - ✓ Keep answers suitable for:
 - 5 marks
 - 8 marks
 - 10 marks
-



Subject:

Social Computing (SC)

? Question:

[Paste your SC question here]



Marks:

[5 / 8 / 10]

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📖 UNIT 3 → Q1 & Q2

What is data mining? Draw and explain architecture of a typical data mining system.

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Here is the SPPU exam-ready answer for **Social Computing (SC)**.

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What is Data Mining? Draw and Explain Architecture of a Typical Data Mining System.

Simple Introduction

Data Mining means finding useful information or hidden patterns from a large amount of data.

It helps companies understand user behavior, trends, and future predictions using stored data.

👉 Simple meaning:

“Finding important knowledge from huge data.”

Definition of Data Mining

Data Mining is the process of extracting useful patterns, relationships, and knowledge from large databases using techniques like analysis, classification, and prediction.

Real-Life Example

Instagram Example

Instagram studies:

- which reels users watch,
- which posts users like,
- what users search,
- how much time users spend.

Using this data, Instagram recommends similar reels and ads.

👉 This is called **Data Mining** because Instagram extracts useful patterns from user data.

Another Social Computing Example

Amazon / Flipkart Recommendation

When you search for shoes:

- website stores your search,
- analyzes your interest,
- recommends similar products.

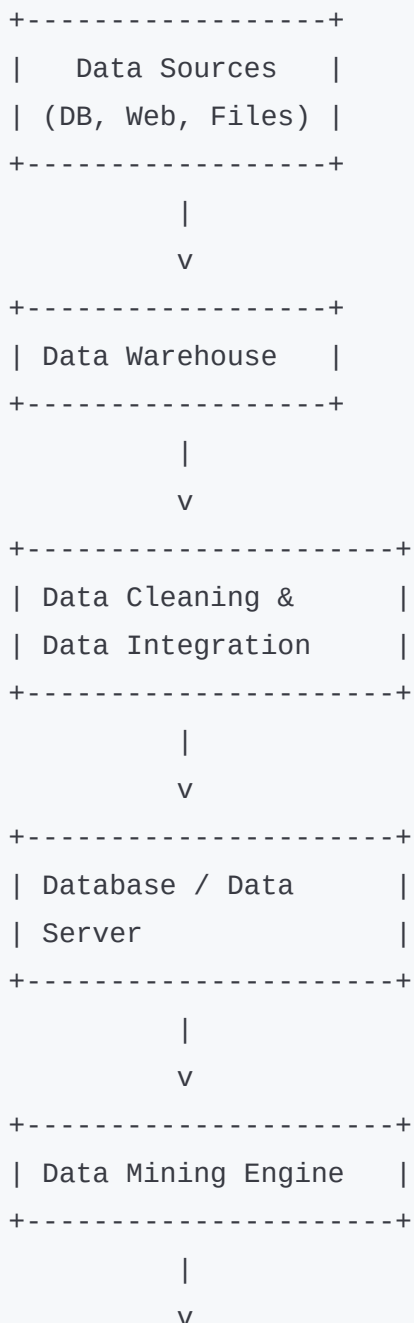
Example:

User searches Shoes → System analyzes data → Recommends Sports Shoes

👉 This is data mining because useful information is extracted from customer data.

✅ Architecture of Typical Data Mining System

📌 Simple Diagram



```
+-----+
| Pattern Evaluation |
+-----+
      |
      v
+-----+
| User Interface / GUI |
+-----+
```

✓ Explanation of Each Component

1 Data Sources

These are original places where data comes from.

Examples:

- databases
- websites
- social media
- files
- online applications

📱 Example:

Instagram user data, YouTube watch history.

2 Data Warehouse

A large storage place where all collected data is stored together.

👉 It stores organized data for analysis.

Example:

- customer records
 - social media activity
 - purchase history
-

3 Data Cleaning and Data Integration

📌 Data Cleaning

Removes:

- duplicate data
- wrong data
- incomplete data

📌 Data Integration

Combines data from many sources into one system.

👉 Clean data gives accurate results.

Example:

Removing fake accounts from Facebook analysis.

4 Database / Data Server

This server manages stored data.

Functions:

- retrieves data,
- updates data,
- sends data to mining engine.

👉 It acts like a manager of the database.

5 Data Mining Engine

This is the main part of the system.

It performs operations like:

- classification
- clustering
- prediction
- association
- trend analysis

👉 It discovers hidden patterns.

Example:

Netflix predicting which movie user may watch next.

6 Pattern Evaluation Module

Checks which patterns are useful and important.

It removes unnecessary patterns.

👉 Only meaningful knowledge is selected.

Example:

Finding most popular hashtags on Twitter/X.

Graphical User Interface (GUI)

Allows users to interact with the system.

Functions:

- enter queries,
- view reports,
- see graphs/charts.

👉 Makes system easy to use.

Example:

Admin dashboard of YouTube analytics.

Working of Data Mining System (Step-by-Step)

Data Collection



Data Storage
↓
Data Cleaning
↓
Pattern Discovery
↓
Pattern Evaluation
↓
Knowledge / Decision

✓ Features of Data Mining

- Extracts hidden information
- Works on huge data
- Helps in prediction
- Supports decision making
- Finds patterns and trends
- Improves business performance

✓ Advantages of Data Mining

Advantages	Explanation
Better decision making	Helps companies take smart decisions
Finds hidden patterns	Discovers useful relationships
Improves marketing	Shows customer interests
Fraud detection	Detects fake activities

Advantages

Explanation

Time saving

Automatic analysis



Disadvantages of Data Mining

Disadvantages

Explanation

Privacy issues

User data may be misused

High cost

Tools are expensive

Complex process

Needs skilled experts

Wrong predictions

Bad data gives wrong results



Applications of Data Mining



Social Media

- Instagram recommendations
- Facebook friend suggestions
- YouTube video recommendations



E-Commerce

- Amazon product suggestions
- Online shopping analysis

Healthcare

- Disease prediction

Banking

- Fraud detection
- Credit card analysis

Education

- Student performance analysis

Memory Trick (Mnemonic)

"SWEEP"

Remember architecture using:

S → Sources

W → Warehouse

E → Engine

E → Evaluation

P → Presentation (GUI)

👉 "SWEEP cleans and finds knowledge from data."

✅ Exam Writing Tips

✅ Underline keywords:

- Data Mining
- Data Warehouse
- Mining Engine
- Pattern Evaluation
- Data Cleaning

✅ Draw architecture diagram properly.

✅ Mention social media examples.

✅ Write advantages/disadvantages in table format.

✅ Use headings for each component.

👉 Diagram + example + keywords = Extra marks in SPPU exam.

✅ Short Conclusion

Data Mining helps in extracting useful knowledge from huge data. It is widely used in social media, business, healthcare, and online platforms for prediction and smart decision making.



Last Minute Revision Points

- Data Mining = extracting useful knowledge from data
 - Main part = Data Mining Engine
 - Data Cleaning removes wrong data
 - Pattern Evaluation selects useful patterns
 - GUI helps user interaction
 - Used in Instagram, YouTube, Amazon
 - Helps prediction and recommendation
 - Important for business analysis
-



Viva Questions with Answers

Q1. What is Data Mining?

Ans: Process of extracting useful information from large data.

Q2. Which component performs mining operations?

Ans: Data Mining Engine.

Q3. What is Data Cleaning?

Ans: Removing incorrect and duplicate data.

Q4. Give one application of Data Mining.



Ans: YouTube video recommendation.

Q5. What is GUI in Data Mining?

Ans: Interface used to interact with the system.

...

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What is classification? Explain with diagram / Explain any one classification algorithm with example.

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What is Classification? Explain with Diagram / Explain Any One Classification Algorithm with Example.

Simple Introduction

Classification is a data mining technique used to divide data into different categories or classes.

It helps in predicting the class or label of new data using old data.

👉 Simple meaning:

“Putting data into correct groups.”

Example:

- Spam or Not Spam email
- Pass or Fail student
- Fake or Real account

Definition of Classification

Classification is the process of finding a model or function that describes and separates data into predefined classes or categories.

Real-Life Example

Instagram Example

Instagram detects:

- fake accounts,
- spam comments,
- real users.

It studies old account behavior and classifies accounts into:

Real Account OR Fake Account

👉 This is classification because data is divided into classes.

Another Social Media Example

Gmail Spam Detection

Gmail checks:

- suspicious links,
- repeated messages,

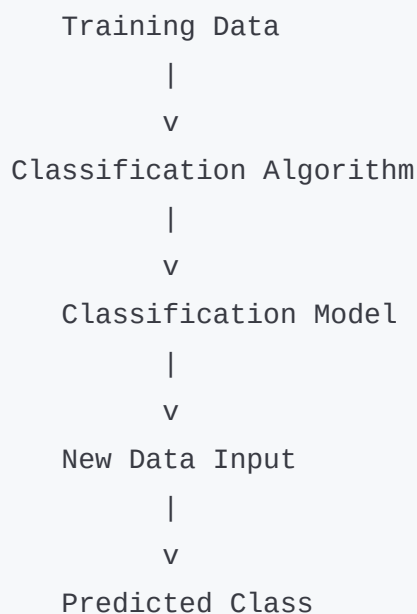
- unknown senders.

Then classifies email into:

Spam Mail OR Normal Mail

👉 This is a classification problem.

✅ Simple Diagram of Classification



✅ Explanation of Diagram

1 Training Data

Old data with labels is given.

Example:

Message	Label
Win money now	Spam
Meeting at 5 PM	Normal

2 Classification Algorithm

Algorithm studies patterns from training data.

Example algorithms:

- Decision Tree
- Naive Bayes
- KNN
- SVM

3 Classification Model

A prediction model is created.

This model learns rules from old data.

4 New Data Input

New unknown data is given to system.

Example:

“Claim free prize now”

Predicted Class

System predicts class.

Example:

Output → Spam

Types of Classification

Type	Meaning
Binary Classification	Two classes
Multi-class Classification	More than two classes

Example

Binary:

- Yes / No
- Spam / Not Spam

Multi-class:

✅ Explain Any One Classification Algorithm

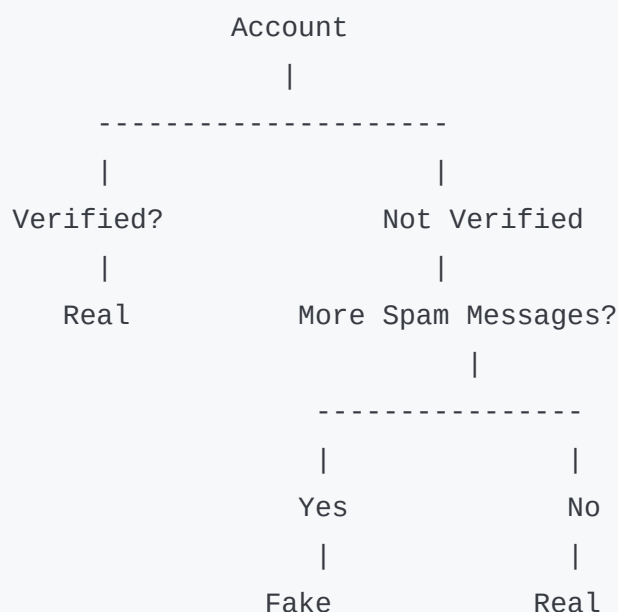
🌳 Decision Tree Algorithm

A **Decision Tree** is a tree-like structure used for classification and decision making.

It asks questions step-by-step and reaches final decision.

✅ Simple Decision Tree Diagram

📱 Example: Classifying Instagram Account



✅ Explanation of Decision Tree

Step 1

Check whether account is verified.

- If verified → Real account

Step 2

If not verified, check spam activity.

- More spam messages → Fake account
- Less spam → Real account

👉 Tree makes decisions step-by-step.

✅ Working of Decision Tree

```
graph TD; A[Input Data] --> B[Ask Question]; B --> C[Split Data]; C --> D[Repeat Process];
```



Final Decision

Features of Classification

- Predicts categories
- Uses historical data
- Easy to understand
- Helps decision making
- Used in machine learning

Advantages of Classification

Advantages	Explanation
Easy prediction	Predicts future classes
Useful in automation	Automatic decision making
High accuracy	Gives good results
Easy understanding	Simple logic

Disadvantages of Classification

Disadvantages	Explanation
Needs training data	Requires old labeled data
Wrong prediction possible	Bad data gives wrong results
Complex for huge data	Large datasets are difficult

Applications of Classification

Social Media

- Fake account detection
- Spam filtering
- Content recommendation

Banking

- Fraud detection
- Loan approval

Healthcare

- Disease prediction

Email Systems

- Spam detection

E-commerce

- Customer category prediction

✅ Real-Life Social Computing Examples

Platform	Classification Example
Instagram	Fake vs Real account
WhatsApp	Spam message detection
YouTube	Video category prediction
Twitter/X	Hate speech detection
LinkedIn	Job recommendation category

✅ Memory Trick (Mnemonic)

📌 "CAP"

C → Categorize

A → Analyze

P → Predict

👉 Classification = Categorize and Predict data.

Exam Writing Tips

 Underline keywords:

- Classification
- Training Data
- Prediction
- Decision Tree
- Classification Model

 Draw decision tree neatly.

 Give social media examples.

 Write advantages/disadvantages in table format.

 Mention real-life applications.

 Diagram + examples + keywords = Extra marks.

Short Conclusion

Classification is an important data mining technique used to predict categories of data. It is widely used in social media, banking, healthcare, and email systems for smart decision making.

Last Minute Revision Points

- Classification divides data into categories

- Uses training data
 - Decision Tree is popular algorithm
 - Used in spam detection
 - Helps prediction
 - Binary classification → 2 classes
 - Multi-class → many categories
 - Social media uses classification heavily
-



Viva Questions with Answers

Q1. What is Classification?

Ans: Process of dividing data into categories.

Q2. Give one example of classification.

Ans: Spam email detection.

Q3. Name one classification algorithm.

Ans: Decision Tree.

Q4. What is training data?

Ans: Old labeled data used for learning.

Q5. Where is classification used in social media?

Ans: Fake account detection.

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Explain Hierarchical clustering algorithm with single linkage clustering.

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Explain Hierarchical Clustering Algorithm with Single Linkage Clustering

Simple Introduction

Hierarchical Clustering is a clustering method used to group similar data items into clusters step-by-step.

It creates a hierarchy (tree-like structure) of clusters.

👉 Simple meaning:

“Making groups from similar data one by one.”

Definition of Hierarchical Clustering

Hierarchical Clustering is a clustering technique in which data objects are grouped into clusters based on similarity, and clusters are formed in hierarchical order.

✅ Real-Life Example



WhatsApp Group Example

Suppose students create groups based on interests:

- Coding students
- Gaming students
- Sports students

First, close friends form small groups.

Then similar groups combine into bigger groups.

👉 This step-by-step grouping is called **Hierarchical Clustering**.

✅ Another Social Media Example



YouTube Recommendation

YouTube groups users based on:

- similar videos watched,
- same interests,
- similar searches.

Example:

Gaming Users → PUBG Fans → Action Game Fans

👉 Similar users are grouped hierarchically.

✓ What is Clustering?

Clustering means grouping similar objects together.

Example:

Similar Users → Same Cluster
Different Users → Different Cluster

✓ Types of Hierarchical Clustering

Type	Meaning
Agglomerative	Bottom to Top
Divisive	Top to Bottom

✓ Agglomerative Hierarchical Clustering

This is the most common method.

👉 Process:

- Start with separate clusters.
- Combine nearest clusters step-by-step.
- Continue until one big cluster is formed.

✓ Simple Diagram of Hierarchical Clustering

```
A   B   C   D
|   |   |   |
A-B  C-D
 \   /
  A-B-C-D
```

👉 Small clusters combine to form bigger clusters.

✓ What is Single Linkage Clustering?

Single Linkage Clustering is a hierarchical clustering method where the distance between two clusters is measured using the **minimum distance** between any two points.

👉 Simple meaning:

“Choose the closest pair of points from two clusters.”

✓ Formula of Single Linkage

```
Distance(C1,C2) =
Minimum distance between points of C1 and C2
```

✓ Simple Diagram of Single Linkage

Cluster 1 Cluster 2

A • ----- • B
Minimum Distance

👉 The smallest distance between clusters is considered.

✓ Working of Hierarchical Clustering using Single Linkage

📌 Step-by-Step Process

Step 1

Treat every data point as separate cluster.

Example:

A B C D

Step 2

Find minimum distance between clusters.

Suppose:

$\text{Distance}(A,B) = \text{Minimum}$

Merge A and B.

(A,B) C D

Step 3

Again find smallest distance.

Suppose:

$\text{Distance}(C,D) = \text{Minimum}$

Merge C and D.

(A,B) (C,D)

Step 4

Find minimum distance between new clusters.

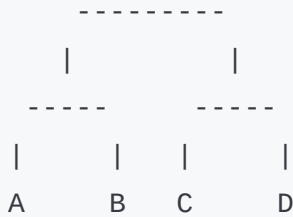
Merge all clusters.

(A,B,C,D)

✅ Dendrogram of Hierarchical Clustering

A **Dendrogram** is a tree diagram showing cluster formation.

📌 Simple Dendrogram



👉 It shows how clusters merge step-by-step.

✅ Features of Hierarchical Clustering

- Creates hierarchy of clusters
- No need to predefine number of clusters
- Easy visualization using dendrogram
- Works well for small datasets

✅ Advantages of Single Linkage Clustering

Advantages	Explanation
Simple method	Easy to understand
Good for similar data	Connects nearest points
No fixed clusters needed	Automatic grouping
Dendrogram visualization	Easy analysis

Disadvantages of Single Linkage Clustering

Disadvantages	Explanation
Sensitive to noise	Wrong data affects result
Chaining problem	Long unwanted clusters form
Slow for large data	High computation time

Applications of Hierarchical Clustering



Social Media

- User community detection
- Friend recommendation



E-Commerce

- Customer grouping

Music Apps

- Similar song grouping

YouTube

- Viewer interest analysis

Healthcare

- Disease classification

Real-Life Social Computing Examples

Platform	Use of Clustering
Instagram	Similar user grouping
Facebook	Community detection
YouTube	Video recommendation
LinkedIn	Professional network grouping
Reddit	Similar discussion communities

Memory Trick (Mnemonic)

"SMILE"

S → Similarity

M → Minimum distance

I → Iterative merging

L → Linkage

E → Expansion of clusters

👉 Single Linkage joins nearest clusters step-by-step.

Exam Writing Tips

 Underline keywords:

- Hierarchical Clustering
- Single Linkage
- Minimum Distance
- Dendrogram
- Agglomerative Method

 Draw dendrogram neatly.

 Mention real-life examples.

 Write advantages/disadvantages in table format.

 Explain steps clearly.

👉 Diagram + steps + example = Extra marks in SPPU exam.

✓ Short Conclusion

Hierarchical clustering groups similar data step-by-step to form clusters. Single linkage clustering uses minimum distance between clusters and is widely used in social media, recommendation systems, and community detection.

✓ Last Minute Revision Points

- Hierarchical clustering creates cluster hierarchy
 - Agglomerative → Bottom to Top
 - Divisive → Top to Bottom
 - Single linkage uses minimum distance
 - Dendrogram shows cluster formation
 - Used in recommendation systems
 - No fixed number of clusters required
 - Sensitive to noise
-

✓ Viva Questions with Answers

Q1. What is Hierarchical Clustering?

Ans: Method of grouping similar data step-by-step.

Q2. What is Single Linkage?

Ans: Method using minimum distance between clusters.

Q3. What is a dendrogram?

Ans: Tree diagram showing cluster merging.

Q4. Name types of hierarchical clustering.

Ans: Agglomerative and Divisive.

Q5. Give one application of clustering.

Ans: YouTube recommendation system.

...

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What is clustering? Explain any clustering algorithm with example.

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What is Clustering? Explain Any Clustering Algorithm with Example.

Simple Introduction

Clustering is a data mining technique used to group similar data items together.

Objects having similar features are placed in the same cluster.

👉 Simple meaning:

"Making groups of similar things."

Example:

- Similar Instagram users in one group
- Similar YouTube videos in one category
- Similar online shoppers in one customer group

Definition of Clustering

Clustering is the process of dividing data objects into groups (clusters) such that objects in the same cluster are more similar to each other than to objects in other clusters.

✅ Real-Life Example

📱 Instagram Example

Instagram groups users based on:

- interests,
- likes,
- reels watched,
- hashtags followed.

Example:

```
Gaming Users → One Cluster  
Fitness Users → One Cluster  
Travel Users → One Cluster
```

👉 Similar users are automatically grouped together.

This is called **Clustering**.

✅ Another Social Computing Example

🎬 YouTube Recommendation

YouTube groups videos based on:

- category,
- watch history,

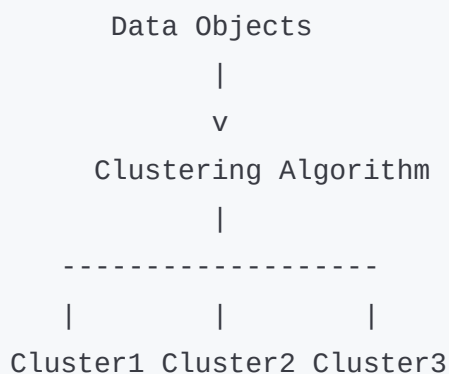
- user interest.

Example:

Action Movies → Cluster 1
Comedy Videos → Cluster 2
Study Videos → Cluster 3

👉 Similar content is grouped together.

✓ Simple Diagram of Clustering



✓ Explanation of Diagram

1 Data Objects

Input data is collected.

Example:

- users,
 - videos,
 - products,
 - posts.
-

Clustering Algorithm

Algorithm checks similarity between objects.

Clusters Formed

Similar objects are grouped together.

Example:

- sports users,
 - music lovers,
 - gamers.
-

Features of Clustering

- Groups similar data
- No predefined labels required
- Finds hidden patterns
- Improves recommendation systems

- Used in machine learning and data mining

✓ Types of Clustering

Type	Meaning
Hierarchical Clustering	Forms clusters step-by-step
Partition Clustering	Divides data into fixed groups
Density-Based Clustering	Forms clusters using density
Grid-Based Clustering	Uses grid structure

✓ Explain Any One Clustering Algorithm

🌟 K-Means Clustering Algorithm

K-Means is one of the most popular clustering algorithms.

It divides data into **K number of clusters**.

👉 "K" means number of groups.

✓ Working of K-Means Clustering



Step-by-Step Process

Step 1 Choose Number of Clusters (K)

Suppose:

$$K = 2$$

Means two clusters are required.

Step 2 Select Initial Centroids

Choose center points randomly.

Cluster 1 Center → C1

Cluster 2 Center → C2

Step 3 Calculate Distance

Each data item checks nearest centroid.

Nearest centroid cluster is assigned.

Step 4 Form Clusters

Similar objects join nearest cluster.



Step 5 Update Centroids

New cluster center is calculated.

Step 6 Repeat Process

Repeat until clusters stop changing.

✓ Simple Diagram of K-Means Clustering



The diagram illustrates the K-Means clustering process. It shows two groups of data points: three black dots on the left and three black triangles on the right. Below the dots is the label 'Cluster1' and below the triangles is the label 'Cluster2'. The text 'Data Points' is centered above the two groups of points.

Data Points

• • • ▲ ▲ ▲

Cluster1 Cluster2

👉 Similar points form one cluster.

✓ Example of K-Means Clustering



Online Shopping Example

An e-commerce website groups customers:

Customer	Interest
A	Mobile Phones
B	Laptops
C	Shoes
D	Clothes

After clustering:

Cluster 1 → Electronics Buyers
Cluster 2 → Fashion Buyers

👉 Customers with similar interests are grouped together.

✅ Real-Life Social Media Examples

Platform	Clustering Example
Instagram	Similar users grouping
YouTube	Video category grouping
Facebook	Community detection
LinkedIn	Professional interest groups
Reddit	Similar discussion communities

✅ Advantages of Clustering

Advantages	Explanation
Finds hidden patterns	Useful insights discovered
Improves recommendations	Better user suggestions
Easy grouping	Similar data together
Helps business analysis	Understands customer behavior

Disadvantages of Clustering

Disadvantages	Explanation
Wrong clusters possible	Bad data affects result
Choosing K is difficult	Incorrect K gives poor result
Sensitive to noise	Outliers affect clusters

Applications of Clustering



Social Media

- Friend suggestions
- Community detection
- Content recommendations



E-Commerce

- Customer segmentation



Healthcare

- Disease grouping



Education

- Student performance grouping



Entertainment

- Movie recommendation systems



Difference Between Classification and Clustering

Classification	Clustering
Uses predefined labels	No predefined labels
Supervised learning	Unsupervised learning
Predicts categories	Groups similar data
Example: Spam detection	Example: Customer grouping



Memory Trick (Mnemonic)



"GROUP"



G → Gather similar data

R → Relate objects

O → Organize clusters

U → Understand patterns

P → Partition data

👉 Clustering groups similar objects.

Exam Writing Tips

✅ Underline keywords:

- Clustering
- K-Means
- Centroid
- Similarity
- Cluster

✅ Draw simple cluster diagram.

✅ Explain K-Means steps clearly.

✅ Mention Instagram/YouTube examples.

✅ Write advantages/disadvantages in table format.

👉 Algorithm steps + diagram + example = Extra marks.

Short Conclusion

Clustering is an important data mining technique used to group similar data together. K-Means clustering is widely used in social media, e-commerce, healthcare, and recommendation systems.

Last Minute Revision Points

- Clustering groups similar data
 - No predefined labels needed
 - K-Means divides data into K clusters
 - Centroid = center of cluster
 - Used in recommendation systems
 - Helps community detection
 - Improves customer analysis
 - Common in Instagram and YouTube
-

Viva Questions with Answers

Q1. What is Clustering?

Ans: Process of grouping similar data together.

Q2. Is clustering supervised learning?

Ans: No, it is unsupervised learning.

Q3. Name one clustering algorithm.

Ans: K-Means algorithm.

Q4. What is centroid?

Ans: Center point of a cluster.

Q5. Give one application of clustering.

Ans: YouTube recommendation system.

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Explain data mining methods for social media.
8

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Explain Data Mining Methods for Social Media

Simple Introduction

Social media platforms generate huge amounts of data every second.

Data Mining Methods for Social Media are techniques used to extract useful information, patterns, trends, and user behavior from social media data.

👉 Simple meaning:

"Finding useful knowledge from social media data."

Definition

Data Mining in Social Media is the process of collecting, analyzing, and extracting meaningful information from platforms like Instagram, Facebook, WhatsApp, YouTube, Twitter/X, LinkedIn, and Reddit.

Why Data Mining is Needed in Social Media?

Social media platforms contain:

- posts,
- comments,
- likes,
- shares,
- videos,
- hashtags,
- chats,
- followers data.

Data mining helps to:

- understand users,
- recommend content,
- detect fake accounts,
- analyze trends,
- improve advertisements.

Simple Diagram

Users → Social Media Data → Data Mining Methods → Useful Information

Explanation

- Users generate data.
 - Social media stores data.
 - Data mining analyzes data.
 - Useful patterns and knowledge are obtained.
-

Data Mining Methods for Social Media

Classification

Classification divides data into categories.

Example

Instagram identifies:

Fake Account OR Real Account

Twitter/X detects:

Spam Tweet OR Normal Tweet

👉 Old data is used to predict category of new data.

Clustering

Clustering groups similar users or content together.

Example

YouTube groups users based on interests:

Gaming Users
Music Lovers
Study Content Users

👉 Similar users are placed in same cluster.

3 Association Rule Mining

Finds relationships between activities or items.

Example

If users watch:

Java Tutorial → DSA Tutorial

Then YouTube recommends DSA videos.

👉 Related patterns are discovered.

4 Sentiment Analysis

Analyzes emotions and opinions in posts/comments.

Example

Twitter/X analyzes tweets:

Positive 😊

Negative 😡

Neutral 😐

Companies use it to know customer opinion.

👉 Helps understand public sentiment.

Trend Analysis

Finds trending topics and viral content.

Example

Twitter/X trending hashtags:

#IPLFinal

#WorldCup

#AI

👉 System analyzes which topics are becoming popular.

6 Recommendation Systems

Suggests content based on user behavior.

Example

Instagram recommends reels based on:

- likes,
- watch history,
- searches,
- follows.

👉 Personalized recommendations improve user experience.

7 Link Analysis

Studies relationships between users or pages.

Example

Facebook friend suggestions:

Mutual Friends → Suggested Friend

LinkedIn also suggests professional connections.

👉 Network relationships are analyzed.

Text Mining

Extracts useful information from text data.

Example

Reddit discussions and YouTube comments are analyzed to understand user interests and opinions.

👉 Important keywords and topics are extracted.

Real-Life Social Media Examples

Platform	Data Mining Method
Instagram	Recommendation system
WhatsApp	Spam detection
Facebook	Friend suggestions
Twitter/X	Sentiment analysis
YouTube	Video recommendation
LinkedIn	Professional connection analysis
Reddit	Community trend analysis
Telegram	Group behavior analysis

✓ Working of Social Media Data Mining

Data Collection



Data Cleaning



Data Analysis



Pattern Discovery



Decision / Recommendation

✓ Features of Social Media Data Mining

- Handles huge data
- Finds hidden patterns
- Predicts user behavior
- Detects fake activities
- Improves user experience
- Supports business decisions

✓ Advantages of Data Mining in Social Media

Advantages	Explanation
Better recommendations	Personalized content
Trend discovery	Finds viral topics
Fraud detection	Detects fake accounts
Business growth	Improves marketing
Customer understanding	Understands user interests

Disadvantages of Data Mining in Social Media

Disadvantages	Explanation
Privacy issues	User data misuse possible
Fake information	Wrong analysis possible
Security risks	Data leakage risk
High computation	Needs powerful systems

Applications / Uses

Social Media Platforms

- Content recommendation
- Spam detection
- Friend suggestion



E-Commerce

- Product recommendation



Banking

- Fraud detection



Healthcare

- Public health analysis



Education

- Student learning analysis



Marketing

- Customer behavior analysis



Memory Trick (Mnemonic)



"CCASTRL"

C → Classification

C → Clustering

A → Association

S → Sentiment Analysis

T → Trend Analysis

R → Recommendation

L → Link Analysis

👉 Helps remember major data mining methods.

Exam Writing Tips

 Underline keywords:

- Classification
- Clustering
- Sentiment Analysis
- Recommendation System
- Trend Analysis

 Draw neat diagram.

 Mention social media examples.

 Write methods in separate headings.

 Use table format for advantages/disadvantages.

👉 Examples from Instagram, YouTube, Twitter/X give extra marks.

Short Conclusion

Data mining methods help social media platforms analyze huge user data and provide useful services like recommendations, trend analysis, spam detection, and community building.

Last Minute Revision Points

- Data mining extracts useful information
 - Social media generates huge data
 - Classification predicts categories
 - Clustering groups similar users
 - Sentiment analysis studies emotions
 - Trend analysis finds viral topics
 - Recommendation systems improve user experience
 - Used in Instagram, YouTube, Facebook, Twitter/X
-

Viva Questions with Answers

Q1. What is data mining?

Ans: Process of extracting useful information from large data.

Q2. Which method groups similar users?

Ans: Clustering.

Q3. What is sentiment analysis?

Ans: Analysis of user emotions and opinions.

Q4. Give one application of data mining in social media.

Ans: YouTube recommendation system.

Q5. What is trend analysis?

Ans: Finding popular and viral topics.

⋮

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Write note on Text mining in social networks.
8

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Write Note on Text Mining in Social Networks



Simple Introduction

Text Mining means extracting useful information from text data.

In social networks, users generate huge amounts of text every day through:

- posts,
- comments,
- chats,
- hashtags,
- tweets,
- reviews,
- messages.

👉 Text mining helps analyze this text and find useful knowledge, opinions, trends, and patterns.



Definition



Text Mining in Social Networks is the process of collecting and analyzing text data from social media platforms to discover useful information, patterns, emotions, and trends.

Real-Life Example

Twitter/X Example

Suppose thousands of users tweet about a movie.

Example tweets:

```
"Movie is amazing!"  
"Worst movie ever!"  
"Songs are fantastic!"
```

Text mining analyzes these tweets and finds:

- positive opinions,
- negative opinions,
- trending topics.

👉 This helps companies understand public reaction.

Another Social Media Example

YouTube Comments Analysis

YouTube analyzes comments like:

"Very helpful tutorial"
"Best explanation"
"Audio quality is poor"

👉 Text mining identifies:

- user feedback,
- user satisfaction,
- common problems.

✅ Simple Diagram of Text Mining

```
graph TD; A["Social Network Text  
(Posts, Comments, Chats)"] --> B["Text Collection"]; B --> C["Text Processing"]; C --> D["Pattern Analysis"]; D --> E["Useful Information / Knowledge"];
```

✅ Explanation of Diagram

1 Social Network Text

Data comes from:

- Instagram captions
 - Twitter tweets
 - Facebook comments
 - Reddit discussions
 - Telegram chats
-

2 Text Collection

The system collects text data from social media.

3 Text Processing

Unwanted words and symbols are removed.

Example:

```
"Movie is very very good!!!"
```

↓

```
"Movie good"
```

4 Pattern Analysis

The system analyzes:

- keywords,

- emotions,
 - trends,
 - user interests.
-

Useful Information

Final results help in:

- decision making,
 - recommendations,
 - advertisements,
 - trend detection.
-

Steps in Text Mining

Step-by-Step Process

```
graph TD; A[Data Collection] --> B[Text Cleaning]; B --> C[Text Transformation]; C --> D[Pattern Discovery]; D --> E[Knowledge Extraction];
```

Data Collection
↓
Text Cleaning
↓
Text Transformation
↓
Pattern Discovery
↓
Knowledge Extraction



Important Techniques Used in Text Mining



Keyword Extraction

Finds important words.



Example

Twitter trending keywords:

```
#IPL  
#AI  
#Election
```



Sentiment Analysis

Detects emotions from text.



Example

Instagram comments:

```
Positive 😊  
Negative 😡  
Neutral 😐
```



3 Topic Detection

Finds main discussion topics.

Example

Reddit communities discussing:

- gaming,
 - coding,
 - fitness,
 - movies.
-

4 Spam Detection

Identifies fake or spam messages.

Example

WhatsApp spam links or fake promotional messages.

5 Recommendation Analysis

Analyzes user interests for recommendations.

Example

YouTube recommends videos based on comments and searches.

Features of Text Mining

- Extracts useful information from text
 - Works on large text data
 - Finds hidden patterns
 - Detects user emotions
 - Supports decision making
 - Helps recommendation systems
-

Applications of Text Mining in Social Networks

Social Media Platforms

- Trend analysis
- Comment analysis
- User behavior analysis

E-Commerce

- Customer review analysis

Marketing

- Brand reputation analysis



Banking

- Fraud message detection



Healthcare

- Public opinion analysis



Education

- Student feedback analysis



Real-Life Social Computing Examples

Platform	Text Mining Use
Instagram	Comment sentiment analysis
Twitter/X	Trend and hashtag analysis
YouTube	Comment recommendation analysis
Facebook	User opinion analysis
Reddit	Topic discussion analysis
Telegram	Spam message detection
LinkedIn	Professional skill trend analysis

Advantages of Text Mining

Advantages	Explanation
Fast analysis	Handles huge text quickly
Finds trends	Detects popular topics
Better recommendations	Understands user interest
Sentiment detection	Identifies opinions
Business improvement	Improves customer service

Disadvantages of Text Mining

Disadvantages	Explanation
Privacy issues	User data misuse risk
Complex language	Slang and emojis difficult
Wrong interpretation	System may misunderstand text
Large data handling	Requires powerful systems

Working Example



Instagram Brand Analysis

Suppose a company launches a new phone.

Users comment:

"Camera is excellent"
"Battery is poor"
"Design looks premium"

Text mining finds:

- positive features,
- customer complaints,
- overall customer opinion.

👉 Company improves product using this analysis.

Memory Trick (Mnemonic)

 "KSTSR"

K → Keyword extraction

S → Sentiment analysis

T → Topic detection

S → Spam detection

R → Recommendation analysis

👉 Helps remember text mining techniques.

Exam Writing Tips

 Underline keywords:

- Text Mining
- Sentiment Analysis
- Keyword Extraction
- Topic Detection
- Spam Detection

 Draw simple text mining diagram.

 Mention examples from Twitter/X, Instagram, YouTube.

 Write applications and advantages in table format.

 Use headings for each technique.

 Social media examples + diagram + keywords = Extra marks.

Short Conclusion

Text mining in social networks helps extract useful information from social media text data. It is widely used for sentiment analysis, trend detection, recommendation systems, and user behavior analysis.

Last Minute Revision Points

- Text mining extracts knowledge from text

- Used in social media platforms
 - Sentiment analysis detects emotions
 - Keyword extraction finds important words
 - Spam detection removes fake messages
 - Helps recommendation systems
 - Used in Instagram, Twitter/X, YouTube
 - Important for trend analysis
-

Viva Questions with Answers

Q1. What is Text Mining?

Ans: Process of extracting useful information from text data.

Q2. What is sentiment analysis?

Ans: Detecting emotions from text.

Q3. Give one application of text mining.

Ans: Twitter trend analysis.

Q4. Which platforms use text mining?

Ans: Instagram, Facebook, YouTube, Twitter/X.

Q5. What is keyword extraction?

Ans: Finding important words from text.

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Explain data representation methods for social media. 8 Nov 2024

★ What is keyword search? Explain query semantics and answer ranking. 9 Apr 2024

★ Explain motivations for data mining in social media. 9 Apr 2024

★ How would you define Twitter? Explain with different services. 8 Nov 2023

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Here are SPPU exam-ready answers for all four questions.

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Explain Data Representation Methods for Social Media



Simple Introduction

Social media generates huge amounts of data like:

- posts,
- likes,
- comments,
- images,
- videos,
- hashtags,
- followers.

To analyze this data properly, it must be represented in organized forms.



👉 These techniques are called **Data Representation Methods**.

✓ Definition

Data Representation in Social Media means organizing and presenting social media data in structured forms so that computers can analyze it easily.

✓ Types of Data Representation Methods

1 Graph Representation

Users and connections are shown as graphs.

```
User A ---- Friend ---- User B
```

👉 Used in Facebook friend networks.

📱 Example

Instagram followers network.

2 Matrix Representation

Data is stored in rows and columns.

User	Likes	Comments
A	20	10
B	15	5

👉 Easy for analysis.

3 Text Representation

Posts, tweets, comments stored as text data.

Example

Twitter/X tweets.

"I love AI technology"

4 Multimedia Representation

Images, videos, reels, audio represented digitally.

Example

YouTube videos and Instagram reels.

5 Tag/Hashtag Representation

Uses hashtags and labels for grouping data.

Example

```
#IPL  
#Travel  
#Fitness
```

👉 Helps trend analysis.

Simple Diagram

```
Social Media Data  
    ↓  
Graph / Matrix / Text / Multimedia  
    ↓  
Analysis and Mining
```

Advantages

- Easy data analysis
- Better recommendation systems
- Trend detection
- Community analysis

Applications

- Facebook friend analysis
 - YouTube recommendation
 - Twitter trend analysis
 - Instagram hashtag analysis
-

Memory Trick

 “GMTTH”

G → Graph

M → Matrix

T → Text

T → Tag

H → Hybrid/Multimedia

Conclusion

Data representation methods organize social media data in useful forms for analysis, recommendation, and trend detection.

Viva Questions

Q1. What is data representation?

Ans: Organizing data into structured forms.

Q2. Which representation uses nodes and links?

Ans: Graph representation.

Q3. Give one example of text representation.

Ans: Twitter tweets.

What is Keyword Search? Explain Query Semantics and Answer Ranking.

Simple Introduction

Keyword Search is a searching method where users enter words to find information.

Example:

Searching:

"Java Tutorial"

on YouTube or Google.

Definition

Keyword Search is a technique where search engines find relevant information based on user-entered keywords.

Working of Keyword Search

```
graph TD; A[User Query] --> B[Search Engine]; B --> C[Matching Data]; C --> D[Ranked Results];
```

Query Semantics

Query semantics means understanding the meaning and intention of user query.

Example

Search:

"Apple"

Possible meanings:

- Apple fruit 🍏
- Apple company 💻

Search engine understands context.

✓ Types of Query Semantics

Type	Example
Exact Meaning	"Java Programming"
Context Meaning	"Apple"
Synonym Search	"Movie" = "Film"

✓ Answer Ranking

Search engine arranges results according to relevance and importance.

Most useful result appears first.

✓ Factors Affecting Ranking

- Keyword match
 - Popularity
 - Likes and shares
 - User engagement
 - Freshness of content
-

✓ Real-Life Example



YouTube Search

Search:

"DSA in Java"

Top videos appear based on:

- views,
- likes,
- watch time,
- relevance.



Advantages

- Fast searching
- Easy information access
- Better user experience



Disadvantages

- Wrong results possible
- Ambiguous keywords
- Spam content



Memory Trick

"QAR"

Q → Query

A → Answer

R → Ranking

Conclusion

Keyword search helps users quickly find information, while query semantics and ranking improve search accuracy and quality.

Viva Questions

Q1. What is keyword search?

Ans: Searching information using keywords.

Q2. What is answer ranking?

Ans: Arranging results by relevance.

Q3. Give one example of keyword search.

Ans: Google search.



Explain Motivations for Data Mining in Social Media



Simple Introduction

Social media creates huge data every second.

Data mining is needed to extract useful information and patterns from this data.

👉 Reasons for using data mining are called **motivations**.



Main Motivations for Data Mining



1 Huge Amount of Data

Millions of users generate data daily.



Example

Instagram reels, YouTube comments, tweets.



2 Understanding User Behavior

Platforms study:

- likes,
- searches,
- comments,



- watch history.

👉 Helps understand user interests.

3 Recommendation Systems

Data mining helps recommend:

- videos,
- reels,
- friends,
- products.

📱 Example

Netflix and YouTube recommendations.

4 Trend Detection

Finds viral topics and hashtags.

📱 Example

Twitter/X trending hashtags.

5 Targeted Advertising

Ads shown according to user interest.

Example

Instagram shopping ads.

6 Fraud and Spam Detection

Detects fake accounts and spam messages.

Example

WhatsApp spam filtering.

7 Business Decision Making

Companies analyze customer opinions.

Example

Twitter customer feedback analysis.

Simple Diagram

Social Media Data



Data Mining



Advantages

- Better recommendations
- Improved marketing
- Trend analysis
- Fraud detection

Conclusion

Data mining is important in social media for analyzing huge data, improving user experience, and helping businesses make smart decisions.

Viva Questions

Q1. Why is data mining needed in social media?

Ans: To analyze huge data and find useful patterns.

Q2. Give one use of data mining.

Ans: Recommendation systems.

Q3. Which platform uses recommendations?

Ans: YouTube.



How Would You Define Twitter? Explain with Different Services.



Simple Introduction

Twitter/X is a social networking and microblogging platform where users share short messages called tweets.

Users can:

- post updates,
- follow people,
- share opinions,
- join discussions.



Definition

Twitter/X is an online social networking platform used for posting short messages, news, opinions, images, and videos in real time.



Features / Services of Twitter/X



1 Tweets

Users post short messages.



Example

"India won the match!"

2 Retweet

Users share others' tweets.

👉 Similar to reposting.

3 Hashtags

Used for topic grouping.

Example

#IPL
#Technology
#AI

4 Followers and Following

Users follow accounts for updates.

Trending Topics

Shows popular discussions.

Direct Messaging (DM)

Private chatting feature.

Spaces

Live audio discussion rooms.

Media Sharing

Users share:

- images,
 - videos,
 - GIFs.
-

Simple Diagram

Advantages of Twitter/X

Advantages	Explanation
Fast news sharing	Real-time updates
Global communication	Connects worldwide users
Trend analysis	Finds popular topics
Easy networking	Follow experts and celebrities

Disadvantages of Twitter/X

Disadvantages	Explanation
Fake news	False information spreads fast
Privacy issues	User data risks
Cyber bullying	Online harassment possible

Applications

- News updates
- Marketing

- Political campaigns
 - Brand promotion
 - Public discussions
-

Memory Trick

"THFMT"

T → Tweets

H → Hashtags

F → Followers

M → Messaging

T → Trending Topics

Conclusion

Twitter/X is a powerful social networking platform used for communication, trend analysis, marketing, and real-time information sharing.

Viva Questions

Q1. What is Twitter/X?

Ans: Social networking and microblogging platform.

Q2. What is a tweet?

Ans: Short message posted on Twitter/X.

Q3. What are hashtags used for?

Ans: Topic grouping and trend analysis.

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UNIT 4 → Q3 & Q4

What is Homophily? Explain how to measure Homophily.
8/9

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What is Homophily? Explain How to Measure Homophily.

Simple Introduction

Homophily means people with similar interests, behavior, background, or opinions tend to connect with each other in social networks.

👉 Simple meaning:

“Similar people become friends.”

Example:

- Gamers join gaming groups
- Coders follow coding pages
- Fitness lovers connect with fitness communities

Definition of Homophily

Homophily is the tendency of individuals to connect and interact with people who are similar to them.

The word means:

Homo = Same
Phily = Love/Attraction

👉 "Love for similar people."

✅ Real-Life Social Media Examples

📱 Instagram Example

People interested in:

- fitness,
- gym,
- healthy food

usually follow similar pages and influencers.

👉 Similar-interest users form a network.

This is **Homophily**.

📱 LinkedIn Example

Software developers mostly connect with:

- programmers,
- recruiters,
- IT professionals.

👉 Professional similarity creates connections.



Facebook Example

College students create groups with classmates having similar interests.

Example:

Coding Group
Gaming Group
Travel Group

👉 Similar users stay connected.



Types of Homophily

Type	Meaning
Status Homophily	Based on age, education, profession
Value Homophily	Based on interests, opinions, hobbies



Status Homophily Example



LinkedIn

IT employees connect with IT employees.

Doctors connect with doctors.

👉 Based on profession.

Value Homophily Example

Instagram

People interested in photography follow photography creators.

👉 Based on interests and values.

Why Homophily Happens?

People prefer similar people because:

- easy communication,
 - common interests,
 - same thinking,
 - trust and comfort,
 - better understanding.
-

Simple Diagram of Homophily

Fitness User — Fitness User

|

└ Gym Community

Gaming User — Gaming User

|

└ Gaming Community

👉 Similar users form social connections.

✓ Measuring Homophily

Homophily is measured by checking how similar connected users are.

✓ Methods to Measure Homophily

1 Similarity Measurement

Checks similarity between connected users.

Example:

- same hobbies,
- same profession,
- same interests.

2 Assortativity Coefficient

Measures similarity in network connections.

Value ranges:

+1 → Strong Homophily
0 → No Homophily
-1 → Opposite Connections

👉 Higher positive value means more similarity.

3 Ratio Measurement

Measures:

Similar Connections

Total Connections

If most friends are similar → High Homophily.

4 Attribute Comparison

Compares user attributes like:

- age,
- gender,

- education,
 - interests,
 - profession.
-

Formula for Homophily Measurement

Simple Formula

Homophily Index =
Similar Connections / Total Connections

Example of Homophily Calculation

Suppose:

- A student has 10 friends.
- 7 friends are from same coding community.

Then:

Homophily Index = $7 / 10 = 0.7$

👉 High homophily because many friends are similar.

✓ Working of Homophily Measurement

```
graph TD; A[User Data] --> B[Check Similarity]; B --> C[Analyze Connections]; C --> D[Calculate Homophily];
```

✓ Features of Homophily

- Similar people connect together
- Creates communities
- Increases interaction
- Common interests improve communication
- Important in social network analysis

✓ Advantages of Homophily

Advantages	Explanation
Better communication	Similar users understand each other
Strong communities	Users stay connected
Easy recommendation	Platforms suggest similar content

Advantages

Explanation

Trust building

Similar people trust more



Disadvantages of Homophily

Disadvantages

Explanation

Limited diversity

Less exposure to new ideas

Echo chambers

Same opinions repeat

Social division

Different groups separate

Bias formation

Similar thinking only



Applications of Homophily



Social Media

- Friend recommendations
- Community detection
- Interest-based groups



E-Commerce

- Product recommendations



Education



- Student communities



LinkedIn

- Professional networking



Entertainment Platforms

- Content recommendation



Real-Life Social Computing Examples

Platform	Homophily Example
Instagram	Fitness communities
Facebook	College groups
LinkedIn	Professional connections
Reddit	Interest-based communities
YouTube	Similar content viewers



Memory Trick (Mnemonic)



"SIMILAR"

S → Same Interests

I → Interaction

M → Matching Users

I → Influence

L → Like-minded

A → Association

R → Relationships

👉 Homophily = Similar people connect together.

Exam Writing Tips

 Underline keywords:

- Homophily
- Similarity
- Social Network
- Assortativity
- Homophily Index

 Draw neat diagram of similar users.

 Mention Instagram and LinkedIn examples.

 Write formula clearly.

 Use table format for advantages/disadvantages.

👉 Formula + examples + diagram = Extra marks in SPPU exam.

✅ Short Conclusion

Homophily is the tendency of similar people to connect in social networks. It helps form communities, improve recommendations, and increase interactions in social media platforms.

✅ Last Minute Revision Points

- Homophily = Similar people connect
 - Common in social media networks
 - Two types → Status and Value
 - Measured using similarity and homophily index
 - Assortativity measures similarity strength
 - Helps recommendation systems
 - Creates communities
 - Used in Instagram, Facebook, LinkedIn
-

✅ Viva Questions with Answers

Q1. What is Homophily?

Ans: Tendency of similar people to connect together.

Q2. Give one example of homophily.

Ans: Gamers joining gaming communities.

Q3. What is status homophily?

Ans: Connection based on profession or education.

Q4. What is homophily index?

Ans: Ratio of similar connections to total connections.

Q5. Which platforms use homophily?

Ans: Instagram, Facebook, LinkedIn.

...

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What is Influence? Explain how to measure influence in social computing.

8/9

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What is Influence? Explain How to Measure Influence in Social Computing.



Simple Introduction

In social computing, some users can affect the opinions, behavior, or decisions of other users.

This effect is called **Influence**.

👉 Simple meaning:

“When one person changes or affects another person’s thinking or action.”

Example:

- Influencers promoting products
- Viral tweets affecting people
- YouTube creators influencing viewers



Definition of Influence



Influence is the ability of a person, group, or social media account to affect the behavior, opinions, decisions, or actions of other users in a social network.

Real-Life Social Media Examples

Instagram Influencer Example

A fitness influencer posts:

"Use this protein powder for muscle growth."

Many followers buy the product.

👉 Followers are influenced by the creator.

YouTube Example

A tech YouTuber reviews a mobile phone.

After watching the video:

- many users buy the phone,
- share the video,
- comment positively.

👉 The YouTuber has influence.



Twitter/X Example

A celebrity tweets about a social issue.

Millions of users:

- like,
- retweet,
- discuss the topic.

👉 Tweet influences public opinion.



Types of Influence

Type	Meaning
Positive Influence	Encourages good actions
Negative Influence	Encourages harmful actions
Direct Influence	Direct communication affects users
Indirect Influence	Influence spreads through others



Positive Influence Example



LinkedIn

Career experts motivate students to learn coding skills.

👉 Helpful and positive influence.



✅ Negative Influence Example

📱 Fake News on Facebook

False information spreads quickly and affects users negatively.

✅ Simple Diagram of Influence

```
graph TD; A[Influencer/User] --> B[Posts / Videos / Tweets]; B --> C[Followers View Content]; C --> D[Likes / Shares / Actions];
```

👉 One user affects many users.

✅ How Influence Spreads in Social Networks

```
graph TD; A[User A] --> B[User B]; B --> C[User C];
```



Community

👉 Influence spreads from one user to many users.

✅ Measuring Influence in Social Computing

Influence is measured by analyzing how much impact a user has on others.

✅ Methods to Measure Influence

1 Number of Followers

More followers usually means higher influence.

📱 Example

Instagram celebrity with 5 million followers.

👉 Large audience = high influence.

2 Likes, Shares, and Comments

Measures engagement of users.

Example:

High Likes + High Shares = Strong Influence

👉 More interaction means stronger influence.

3 Retweet/Repost Count

Measures how much content spreads.



Example

Twitter/X viral tweet shared by thousands.

👉 High reposts indicate strong influence.

4 PageRank / Network Centrality

Measures importance of user in social network.

Users connected to many important users have higher influence.

5 Engagement Rate

Measures user interaction level.



Formula

Engagement Rate =
(Likes + Comments + Shares) / Followers

👉 Higher engagement = more influence.

6 Content Reach

Measures how many users saw the content.

Example:

- Instagram reel views
- YouTube video reach

7 Viral Spread Analysis

Measures how quickly content spreads across network.

✅ Example of Influence Measurement

Suppose:

User	Followers	Likes
A	1 Million	50K
B	5K	100

👉 User A has higher influence.

✅ Influence Measurement Process

```
graph TD; A[Social Media Data] --> B[Analyze User Activity]; B --> C[Measure Engagement]; C --> D[Calculate Influence];
```

✅ Features of Influence

- Affects user behavior
- Spreads information quickly
- Helps viral marketing
- Important in social networks
- Measured using engagement and reach

✓ Advantages of Influence in Social Computing

Advantages	Explanation
Fast information spread	News spreads quickly
Helpful marketing	Promotes products
Community awareness	Social campaigns become popular
Better communication	Connects users globally

✓ Disadvantages of Influence

Disadvantages	Explanation
Fake news spread	Wrong information spreads fast
Negative influence	Harmful trends may spread
Manipulation	Users can be misled
Privacy concerns	User data misuse possible

✓ Applications of Influence Measurement



Social Media Marketing

Brands identify influencers for promotion.

E-Commerce

Influencer marketing increases sales.

Politics

Political campaigns use influential users.

Public Awareness

Awareness campaigns spread through influencers.

Entertainment

Movie promotions by celebrities.

Real-Life Social Computing Examples

Platform	Influence Example
Instagram	Influencer marketing
YouTube	Tech reviewers
Twitter/X	Viral tweets
LinkedIn	Career influencers
Facebook	Viral awareness campaigns



Memory Trick (Mnemonic)



"FLARES"

F → Followers

L → Likes

A → Actions

R → Reach

E → Engagement

S → Shares

👉 These factors help measure influence.



Exam Writing Tips

✓ Underline keywords:

- Influence
- Engagement Rate
- Followers
- Reach
- Viral Spread

✓ Draw influence flow diagram.



✅ Mention Instagram and Twitter/X examples.

✅ Write formulas clearly.

✅ Use tables for advantages/disadvantages.

👉 Diagram + formula + examples = Extra marks.

✅ Short Conclusion

Influence in social computing refers to the ability of users to affect others in social networks. It is measured using followers, engagement, shares, and content reach, and is widely used in marketing and recommendation systems.

✅ Last Minute Revision Points

- Influence affects user behavior
 - Influencers spread information
 - Measured using followers and engagement
 - Engagement rate is important
 - Viral content shows strong influence
 - Used in marketing and awareness campaigns
 - Common on Instagram, YouTube, Twitter/X
 - Can be positive or negative
-



Viva Questions with Answers

Q1. What is influence in social computing?

Ans: Ability to affect user behavior and opinions.

Q2. Give one example of influence.

Ans: Instagram influencer marketing.

Q3. What measures influence?

Ans: Followers, likes, shares, engagement.

Q4. What is engagement rate?

Ans: Measure of user interaction.

Q5. Which platform uses influencers heavily?

Ans: Instagram.

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Explain Shuffle Test and Edge Reversal Test.
9

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Explain Shuffle Test and Edge Reversal Test



Simple Introduction

In social networks, researchers study whether users influence each other or simply connect because they are similar.

To analyze this, special testing methods are used:

- Shuffle Test
- Edge Reversal Test

These methods help understand:

- influence,
- homophily,
- social behavior.

👉 Simple meaning:

“These tests check whether users truly influence each other in social networks.”



Why These Tests are Needed?



In social media:

- friends may behave similarly because of influence,
- or because they already have similar interests.

Example:

Two Instagram friends both like fitness content.

Question:

Did one friend influence the other?
OR
Were both already interested in fitness?

👉 Shuffle Test and Edge Reversal Test help answer this.

1 Shuffle Test

Definition

Shuffle Test is a method used to check whether similarity between users happens because of actual social influence or only by chance.

👉 It randomly shuffles user data and compares results.

Working of Shuffle Test

Step-by-Step Process

Step 1

Collect social network data.

Example:

- Instagram friends,
 - YouTube viewers,
 - Twitter followers.
-

Step 2

Observe actual behavior similarity.

Example:

Friends watching same reels.

Step 3

Randomly shuffle user behaviors.

Connections remain same but behavior data changes randomly.

Step 4

Compare original and shuffled results.

- If similarity decreases after shuffling → real influence exists.

- If similarity remains same → similarity happened by chance.

✓ Simple Diagram of Shuffle Test

Original Network
(User A → Fitness Content)

↓ Shuffle Behavior

Randomized Network
(User A → Random Content)

↓

Compare Results

✓ Real-Life Example of Shuffle Test



Instagram Example

Suppose:

- many friends watch same fitness reels.

System shuffles viewing behavior randomly.

If similarity disappears after shuffle:

👉 Friends were actually influencing each other.

✓ Advantages of Shuffle Test

Advantages

Explanation

Detects real influence	Separates chance from influence
Simple statistical method	Easy analysis
Useful in social network analysis	Widely used in research

✓ Disadvantages of Shuffle Test

Disadvantages

Explanation

Computationally expensive	Needs repeated shuffling
Complex for huge networks	Large data handling difficult



2 Edge Reversal Test

✓ Definition

Edge Reversal Test checks influence direction by reversing relationships between users in a social network.

👉 It studies whether influence depends on direction of connection.

✓ Meaning of Edge

In social networks:

User A → User B

This connection is called an **edge**.

Working of Edge Reversal Test

Step-by-Step Process

Step 1

Study original social connections.

Example:

A follows B

Step 2

Measure influence in original direction.

Step 3

Reverse connection direction.

B follows A

Step 4

Measure influence again.

Step 5

Compare both results.

- If influence changes strongly → actual influence exists.
 - If no change → similarity may be due to homophily.
-

Simple Diagram of Edge Reversal Test

Original Edge:

A → B

Reversed Edge:

B → A

👉 Influence is checked in both directions.

Real-Life Example of Edge Reversal Test

Twitter/X Example

Suppose:

- User A follows celebrity B.
- User A starts using same hashtags as B.

Now reverse edge:

Celebrity B follows User A

If influence pattern changes:

👉 Original direction had real influence.

Advantages of Edge Reversal Test

Advantages	Explanation
Detects influence direction	Checks who influences whom
Separates homophily and influence	Better social analysis
Useful for network research	Popular in social computing

Disadvantages of Edge Reversal Test

Disadvantages

Explanation

Complex network analysis

Difficult in huge networks

Direction may not always matter

Some networks are symmetric

Difference Between Shuffle Test and Edge Reversal Test

Shuffle Test

Edge Reversal Test

Randomly shuffles behavior

Reverses network edges

Checks randomness

Checks direction of influence

Focuses on behavior similarity

Focuses on connection direction

Used for influence validation

Used for direction analysis

Applications of These Tests



Social Media

- Instagram influence analysis
- Twitter trend spread analysis
- Facebook community studies

Marketing

- Influencer impact analysis

Research

- Social behavior analysis

E-Commerce

- Product recommendation influence

Real-Life Social Computing Examples

Platform	Example
Instagram	Reel influence analysis
Twitter/X	Viral hashtag spread
Facebook	Friend behavior similarity
YouTube	Content recommendation influence
LinkedIn	Professional influence analysis

Memory Trick (Mnemonic)

 "SER"

S → Shuffle

E → Edge reversal

R → Relationship analysis

👉 Used to study social influence.

Exam Writing Tips

 Underline keywords:

- Shuffle Test
- Edge Reversal Test
- Influence
- Homophily
- Social Network

 Draw both diagrams neatly.

 Mention Instagram and Twitter/X examples.

 Write comparison table.

 Explain step-by-step working.

👉 Diagrams + comparison + examples = Extra marks.

Short Conclusion

Shuffle Test and Edge Reversal Test are important methods used in social computing to study social influence and distinguish it from homophily. These methods help analyze user behavior in social networks accurately.

Last Minute Revision Points

- Shuffle Test randomizes behavior
 - Edge Reversal reverses connections
 - Both study social influence
 - Helps separate influence from homophily
 - Used in social network analysis
 - Important in Instagram and Twitter research
 - Shuffle → randomness check
 - Edge reversal → direction check
-

Viva Questions with Answers

Q1. What is Shuffle Test?

Ans: Method that randomizes behavior to check influence.

Q2. What is Edge Reversal Test?

Ans: Method that reverses network direction to analyze influence.

Q3. Why are these tests used?

Ans: To study social influence and homophily.

Q4. What is an edge in social network?

Ans: Connection between users.

Q5. Which platforms use influence analysis?

Ans: Instagram, Twitter/X, Facebook.

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Explain Randomization Test with example / in detail.
8/9

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Explain Randomization Test with Example / In Detail

Simple Introduction

In social computing and data analysis, researchers need to check whether a result happened:

- due to real influence,
OR
- only by chance.

A **Randomization Test** helps verify this.

👉 Simple meaning:

"Randomization Test checks whether a result is real or random."

Definition

Randomization Test is a statistical testing method where data values are randomly rearranged many times to check whether observed results are significant or happened by chance.



Why Randomization Test is Used?

It is used to:

- check social influence,
- test hypotheses,
- analyze user behavior,
- validate data mining results.

👉 Helps determine whether patterns are meaningful.



Real-Life Social Media Example



Instagram Example

Suppose researchers observe:

Friends watch same fitness reels.

Question:

Are friends influencing each other?

OR

Did similarity happen randomly?

👉 Randomization Test helps answer this.



✓ Another Example

📱 Twitter/X Example

Suppose a hashtag becomes viral.

Researchers test:

Did users spread hashtag because of influence?
OR
Did it spread randomly?

👉 Randomization Test analyzes this.

✓ Working of Randomization Test

📌 Step-by-Step Process

Step 1 Collect Original Data

Collect social network behavior.

Example:

- likes,
 - shares,
 - follows,
 - comments.
-

Step 2 Calculate Original Result

Find actual similarity or influence.

Example:

80% friends like same reels

Step 3 Randomize Data

Shuffle data randomly many times.

Connections stay same but behavior changes randomly.

Step 4 Recalculate Result

After randomization, calculate similarity again.

Example:

Random similarity = 20%

Step 5 Compare Results

If original result is much higher than randomized result:

👉 Real influence exists.

Otherwise:

👉 Similarity may be random.

✅ Simple Diagram of Randomization Test

```
graph TD; A[Original Social Data] --> B[Measure Actual Result]; B --> C[Randomize Data]; C --> D[Measure Random Result]; D --> E[Compare Both Results];
```

✅ Detailed Example of Randomization Test



Facebook Example

Suppose:

- 100 friends use same hashtag.
- Researchers suspect social influence.

Original Observation

Same Hashtag Usage = 75%

After Randomization

Behavior shuffled randomly.

Result becomes:

Random Similarity = 25%

Final Conclusion

Because original similarity is much higher:

👉 Real social influence exists.

Important Terms in Randomization Test

Term	Meaning
Original Data	Actual collected data
Randomized Data	Shuffled data

Term	Meaning
Significance	Importance of result
Similarity	Common behavior between users

Applications of Randomization Test



Social Media

- Influence analysis
- Trend analysis
- Community behavior study



Marketing

- Advertisement effectiveness



Research

- Social behavior analysis



E-Commerce

- Product recommendation validation

Features of Randomization Test

- Statistical testing method
- Uses random shuffling
- Checks result significance
- Detects real influence
- Useful in network analysis

Advantages of Randomization Test

Advantages	Explanation
Detects real patterns	Separates randomness from influence
Flexible method	Works on many datasets
No strict assumptions	Easy for social data
Accurate validation	Improves analysis quality

Disadvantages of Randomization Test

Disadvantages	Explanation
High computation cost	Many calculations needed
Slow for huge data	Large networks difficult
Complex implementation	Hard for beginners

Randomization Test vs Shuffle Test

Randomization Test

General statistical method

Used in many analyses

Randomizes dataset

Broader concept

Shuffle Test

Specific random shuffling method

Mainly for influence checking

Randomizes behavior data

Part of randomization techniques

Real-Life Social Computing Examples

Platform

Example

Instagram

Reel influence analysis

Facebook

Hashtag spread analysis

Twitter/X

Viral trend checking

YouTube

Recommendation influence

LinkedIn

Professional influence analysis

Memory Trick (Mnemonic)



"RANDOM"

R → Rearrange data

A → Analyze again

N → Notice differences

D → Detect significance

O → Observe influence

M → Measure results

👉 Randomization checks whether results are real.

Exam Writing Tips

 Underline keywords:

- Randomization Test
- Significance
- Random Data
- Social Influence
- Similarity

 Draw neat process diagram.

 Mention Instagram/Twitter examples.

 Write step-by-step working clearly.

 Use tables for advantages/disadvantages.

👉 Example + steps + diagram = Extra marks.

Short Conclusion

Randomization Test is an important statistical method used in social computing to determine whether social behavior patterns are real or occurred randomly. It helps analyze influence, trends, and user interactions accurately.

Last Minute Revision Points

- Randomization Test checks significance
 - Data is shuffled randomly
 - Compares original and randomized results
 - Used in social influence analysis
 - Detects real patterns
 - Helpful in social network analysis
 - Used in Instagram and Twitter research
 - Important for data validation
-

Viva Questions with Answers

Q1. What is Randomization Test?

Ans: Statistical method to check whether results are real or random.

Q2. Why is Randomization Test used?

Ans: To analyze significance of patterns.

Q3. What happens during randomization?

Ans: Data values are shuffled randomly.

Q4. Give one application of Randomization Test.

Ans: Social influence analysis.

Q5. Which platforms use influence analysis?

Ans: Instagram, Facebook, Twitter/X.

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What is Assortativity? Explain any one technique to measure assortativity.

9

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What is Assortativity? Explain Any One Technique to Measure Assortativity.

Simple Introduction

In social networks, people usually connect with others who are similar to them.

This similarity-based connection pattern is called **Assortativity**.

👉 Simple meaning:

“Similar people connect with similar people.”

Example:

- Gamers follow gamers
- Doctors connect with doctors
- Students join student groups

Definition of Assortativity

Assortativity is a measure used in social network analysis to check whether nodes (users) in a network connect with similar or different nodes.

👉 It measures similarity in connections.

Real-Life Social Media Examples

LinkedIn Example

Software engineers mostly connect with:

- programmers,
- IT companies,
- recruiters.

👉 Similar profession-based connections show assortativity.

Instagram Example

Fitness lovers mostly follow:

- gym pages,
- fitness influencers,
- diet channels.

👉 Similar-interest users connect together.



Facebook Example

College students form groups with classmates having similar hobbies and education.

👉 Similarity-based grouping is assortativity.



Types of Assortativity

Type	Meaning
Positive Assortativity	Similar users connect together
Negative Assortativity	Different users connect together



Positive Assortativity Example

Gamers → Connect with Gamers

👉 Similar people connect.



Negative Assortativity Example

Teacher → Connect with Students

👉 Different types connect.

✅ Simple Diagram of Assortativity

```
graph TD
    A[Fitness User] --- B[Fitness User]
    A --- C[Gym Community]
    D[Coder] --- E[Programmer]
```

👉 Similar users form network connections.

✅ Why Assortativity is Important?

It helps to:

- analyze social networks,
 - study community formation,
 - detect influence patterns,
 - improve recommendation systems.
-

✅ Technique to Measure Assortativity

🌟 Assortativity Coefficient Technique

The most common method used to measure assortativity is the **Assortativity Coefficient**.

✓ Assortativity Coefficient

It measures similarity between connected users in a network.

✓ Range of Assortativity Coefficient

+1 → Strong Positive Assortativity
0 → No Assortativity
-1 → Negative Assortativity

✓ Meaning of Values

Value	Meaning
+1	Highly similar connections
0	Random connections
-1	Opposite-type connections

✓ Formula of Assortativity Coefficient

📌 Simple Formula

$r =$
Similarity between connected nodes

Total possible similarity

👉 Higher value means stronger similarity.

✅ Working of Assortativity Measurement

📌 Step-by-Step Process

Step 1 Collect Network Data

Collect user connection data.

Example:

- followers,
- friends,
- communities.

Step 2 Identify User Attributes

Check similarities like:

- age,
- profession,
- interests,
- education.

Step 3 Compare Connected Users

Measure how many connected users are similar.

Step 4 Calculate Coefficient

Find assortativity value.

Step 5 Analyze Result

- Positive value → Similar users connect.
 - Negative value → Different users connect.
-

Example of Assortativity Measurement

Suppose:

User	Profession
A	Engineer
B	Engineer
C	Doctor

Connections:

A ↔ B

Both are engineers.

👉 High positive assortativity.

✓ Another Example



LinkedIn

If most IT professionals connect with IT professionals:

Assortativity = Positive

✓ Applications of Assortativity



Social Media

- Community detection
- Friend recommendation



Professional Networks

- LinkedIn connection analysis



E-Commerce

- Customer behavior analysis



Education Platforms

- Student group analysis



Entertainment Platforms

- Similar audience grouping



Features of Assortativity

- Measures network similarity
- Detects community structure
- Useful in social network analysis
- Helps recommendation systems
- Identifies user behavior patterns



Advantages of Assortativity

Advantages	Explanation
Finds similar communities	Easy community detection
Improves recommendations	Suggests relevant users/content



Advantages	Explanation
Helps network analysis	Understands social behavior
Useful in research	Studies social structures

Disadvantages of Assortativity

Disadvantages	Explanation
Complex for huge networks	Difficult calculations
Limited diversity	Similar groups only
May create echo chambers	Same opinions repeat

Difference Between Homophily and Assortativity

Homophily	Assortativity
Tendency to connect with similar users	Measurement of similarity
Social behavior concept	Mathematical/network measure
Explains why users connect	Measures how strongly users connect

Real-Life Social Computing Examples

Platform	Assortativity Example
Instagram	Fitness communities
LinkedIn	Professional networking
Facebook	Student groups
Reddit	Interest-based communities
YouTube	Similar audience groups

✅ Memory Trick (Mnemonic)

📌 "SIMILAR"

S → Same Interests

I → Interaction

M → Matching Users

I → Influence

L → Link Similarity

A → Association

R → Relationships

👉 Assortativity measures similarity in relationships.

Exam Writing Tips

 Underline keywords:

- Assortativity
- Assortativity Coefficient
- Similarity
- Positive Assortativity
- Network Analysis

 Draw network diagram neatly.

 Mention LinkedIn and Instagram examples.

 Write coefficient range clearly.

 Use table format wherever possible.

 Formula + examples + diagram = Extra marks.

Short Conclusion

Assortativity is an important concept in social computing used to measure similarity between connected users in social networks. It helps analyze communities, recommendations, and user behavior patterns.

Last Minute Revision Points

- Assortativity measures similarity

- Positive assortativity → similar users connect
 - Negative assortativity → different users connect
 - Measured using assortativity coefficient
 - Important in social network analysis
 - Helps recommendation systems
 - Common in LinkedIn and Instagram
 - Related to homophily
-

Viva Questions with Answers

Q1. What is assortativity?

Ans: Measure of similarity between connected users.

Q2. What is positive assortativity?

Ans: Similar users connect together.

Q3. What is assortativity coefficient?

Ans: Numerical measure of network similarity.

Q4. Give one example of assortativity.

Ans: Engineers connecting with engineers on LinkedIn.

Q5. Which platforms use assortativity analysis?

Ans: Instagram, LinkedIn, Facebook.

...

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Distinguish Influence and Homophily. 9 Apr 2023

★ What is a shuffle test designed for in the context of social influence? Describe how it is performed. 9 Nov 2023

★ Describe how the edge reversal test works. What is it used for? 9

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Distinguish Between Influence and Homophily

Simple Introduction

In social networks, users often behave similarly.

This similarity may happen because:

- users influence each other,
OR
- users already have similar interests.

These concepts are called:

- Influence
- Homophily

Definition of Influence

Influence means one user changes or affects another user's behavior, opinion, or action.

👉 Example:

Instagram influencer promotes shoes and followers buy them.

✅ Definition of Homophily

Homophily means similar people naturally connect together because they already share common interests or characteristics.

👉 Example:

Gamers joining gaming communities.

✅ Difference Between Influence and Homophily

Influence	Homophily
Users affect others' behavior	Similar users connect naturally
Behavior changes after connection	Similarity exists before connection
Based on social effect	Based on common interests
Creates behavioral changes	Creates similar communities
Example: Influencer marketing	Example: Fitness groups
Dynamic process	Natural similarity process
Focuses on action spreading	Focuses on relationship formation

✅ Real-Life Example

📱 Instagram Example

Influence

A fitness influencer posts workout reels.

Followers start exercising.

👉 Followers changed behavior because of influence.

Homophily

Fitness lovers follow fitness pages because they already like fitness.

👉 Similar-interest users connected naturally.

✅ Simple Diagram

Influence:

Influencer → Followers → Behavior Change

Homophily:

Similar Users → Connection → Community

✓ Features of Influence

- Behavior spreading
 - Opinion change
 - Viral content sharing
 - Marketing impact
-

✓ Features of Homophily

- Similar people connect
 - Common interests
 - Community formation
 - Similar behavior patterns
-

✓ Applications

Influence	Homophily
Influencer marketing	Friend recommendation
Viral advertising	Community detection
Political campaigns	Interest-based grouping

✓ Memory Trick

"ICH"

I → Influence changes behavior

C → Connection through similarity

H → Homophily means similar people

Conclusion

Influence changes user behavior, while homophily explains why similar users connect together in social networks.

Viva Questions

Q1. What is influence?

Ans: Ability to affect user behavior.

Q2. What is homophily?

Ans: Similar people connecting together.

Q3. Give one example of influence.

Ans: Influencer marketing.

 **What is a Shuffle Test Designed for in the Context of Social Influence?**

Describe How it is Performed.

Simple Introduction

In social networks, researchers need to check whether user behavior similarity happens because of:

- real social influence,
OR
- random chance.

👉 Shuffle Test is used for this purpose.

Definition

Shuffle Test is a statistical testing method used to determine whether observed similarity between users is caused by social influence or occurred randomly.

Purpose of Shuffle Test

Shuffle Test is designed to:

- detect real social influence,
 - separate randomness from actual influence,
 - analyze social behavior.
-

Real-Life Example

Instagram Example

Friends watch same fitness reels.

Question:

Are friends influencing each other?
OR
Did similarity happen by chance?

👉 Shuffle Test helps answer this.

How Shuffle Test is Performed

Step-by-Step Process

Step Collect Original Data

Collect user behavior data.

Example:

- likes,
 - comments,
 - views,
 - shares.
-

Step 2 Measure Actual Similarity

Find how similar connected users are.

Example:

80% friends like same reels

Step 3 Shuffle User Behavior Randomly

Randomly rearrange behavior data.

Connections remain same.

Step 4 Recalculate Similarity

Find similarity after randomization.

Example:

Random similarity = 20%

Step 5 Compare Results

If original similarity is much higher:

👉 Real social influence exists.

Otherwise:

👉 Similarity may be random.

✓ Simple Diagram

```
graph TD; A[Original Data] --> B[Measure Similarity]; B --> C[Shuffle Data]; C --> D[Measure Again]; D --> E[Compare Results];
```

✓ Applications

- Influence analysis
 - Viral content study
 - Social network research
 - Recommendation systems
-

✓ Advantages

Advantages	Explanation
Detects real influence	Removes randomness
Easy statistical analysis	Useful in social computing
Supports behavior study	Better network understanding

Conclusion

Shuffle Test is an important method used to verify whether social influence in networks is real or only random.

Viva Questions

Q1. What is Shuffle Test?

Ans: Method used to detect real social influence.

Q2. Why is data shuffled?

Ans: To remove random effects.

Q3. Where is Shuffle Test used?

Ans: Social network analysis.

Describe How the Edge Reversal Test Works. What is it Used For?

✓ Simple Introduction

In social networks, users influence each other through connections.

The **Edge Reversal Test** checks whether influence depends on direction of connection.

👉 It studies who influences whom.

✓ Definition

Edge Reversal Test is a method used to analyze social influence by reversing the direction of connections (edges) in a social network.

✓ Meaning of Edge

In social networks:

$A \rightarrow B$

This connection is called an **edge**.

✓ Purpose of Edge Reversal Test

Used to:

- study direction of influence,

- distinguish influence from homophily,
 - analyze network behavior.
-

Working of Edge Reversal Test

Step-by-Step Process

Step **1** Observe Original Connection

Example:

User A follows User B

Step **2** Measure Influence

Check how User B affects User A.

Step **3** Reverse Edge Direction

User B follows User A

Step **4** Measure Influence Again

Analyze changes in influence pattern.

Step **5** Compare Results

- Major change → real influence exists.
 - No change → similarity may be due to homophily.
-

Simple Diagram

Original:

$A \rightarrow B$

Reversed:

$B \rightarrow A$

Real-Life Example



Twitter/X Example

A celebrity posts hashtags.

Followers start using same hashtags.

Now reverse relationship:

Celebrity follows followers

If influence changes:

👉 Original direction had strong influence.

✓ Applications

- Social influence analysis
- Viral marketing study
- Recommendation systems
- Community behavior analysis

✓ Advantages

Advantages

Explanation

Detects influence direction

Finds who influences whom

Separates homophily

Better network analysis

Useful in research

Important in social computing

✓ Disadvantages

Disadvantages

Explanation

Complex analysis

Difficult for huge networks

Computationally expensive

Needs large calculations

Memory Trick

"REV"

R → Reverse edge

E → Evaluate influence

V → Verify direction

Conclusion

Edge Reversal Test is used in social computing to analyze the direction of social influence by reversing user connections and comparing influence patterns.

Viva Questions

Q1. What is Edge Reversal Test?

Ans: Method that reverses network connections to study influence.

Q2. What is an edge?

Ans: Connection between users.

Q3. Why is Edge Reversal Test used?

Ans: To study influence direction.

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📖 UNIT 5 → Q5 & Q6

☆☆☆ What is Individual Behavior? Explain individual online behavior three categories. 8/9

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What is Individual Behavior? Explain Individual Online Behavior Three Categories.

Simple Introduction

In social computing, every user behaves differently on online platforms.

The way a person:

- communicates,
- posts,
- likes,
- comments,
- searches,
- reacts online

is called **Individual Behavior**.

👉 Simple meaning:

“How a person acts on the internet.”

Definition of Individual Behavior

Individual Behavior refers to the actions, reactions, and activities performed by a user on online platforms and social networks.

It studies how a person behaves while using:

- Instagram,
- WhatsApp,
- YouTube,
- Facebook,
- Twitter/X,
- LinkedIn,
- online communities.

Real-Life Example



Instagram Example

Different users behave differently:

User Type	Behavior
User A	Watches reels only
User B	Uploads photos daily
User C	Comments on every post

👉 These actions represent individual behavior.

✓ Why Study Individual Behavior?

It helps platforms to:

- recommend content,
 - show advertisements,
 - improve user experience,
 - detect fake accounts,
 - understand user interests.
-

✓ Categories of Individual Online Behavior

Individual online behavior is mainly divided into **three categories**:

- 1 Information Consumption Behavior
 - 2 Information Participation Behavior
 - 3 Information Production Behavior
-

1 Information Consumption Behavior

✓ Meaning

Users consume or view information online without creating content.

👉 User mainly reads, watches, or listens.

✅ Activities

- Watching YouTube videos
 - Reading Facebook posts
 - Watching Instagram reels
 - Reading tweets on Twitter/X
-

✅ Real-Life Example

📱 YouTube Example

A user watches:

- coding tutorials,
- gaming videos,
- music videos

but never uploads or comments.

👉 This is **Information Consumption Behavior**.

✅ Simple Diagram

Social Media Content



User Views



Information Consumption



Features

- Passive behavior
- User only consumes data
- No content creation
- Common in most users



Information Participation Behavior



Meaning

Users interact with content created by others.

👉 Users participate but may not create original content.



Activities



- Liking posts
 - Sharing reels
 - Commenting on videos
 - Retweeting tweets
-

✓ Real-Life Example

📱 Instagram Example

A user:

- likes reels,
- comments on posts,
- shares memes with friends.

👉 This is **Participation Behavior**.

✓ Simple Diagram

```
graph TD; User --> Actions[Likes / Comments / Shares]; Actions --> Participation;
```

Features

- Active interaction
 - Improves engagement
 - Helps viral content spread
 - Creates communication
-

Information Production Behavior

Meaning

Users create and upload original content online.

👉 User becomes content creator.

Activities

- Uploading YouTube videos
 - Posting Instagram reels
 - Writing tweets
 - Creating blogs
-

✓ Real-Life Example



YouTube Creator Example

A teacher uploads:

- Java tutorials,
- DSA lectures,
- coding videos.

👉 This is **Information Production Behavior**.

✓ Simple Diagram

```
graph TD; A[User Creates Content] --> B[Upload to Social Platform]; B --> C[Audience Views Content];
```

✓ Features

- Creative behavior
- Produces original content
- Influences other users
- Important for social media growth

✓ Comparison of Three Categories

Category	Main Activity	Example
Consumption	Viewing content	Watching reels
Participation	Interacting with content	Liking/commenting
Production	Creating content	Uploading videos

✓ Working of Individual Online Behavior

```
User Activity
  ↓
Consumption / Participation / Production
  ↓
Online Behavior Analysis
```

✓ Applications of Individual Behavior Analysis



Social Media Platforms

- Personalized recommendations
- User engagement analysis



Marketing

- Targeted advertisements



E-Commerce

- Product recommendations



Education Platforms

- Student learning analysis



Entertainment Platforms

- Movie and music suggestions



Advantages of Studying Individual Behavior

Advantages	Explanation
Better recommendations	Personalized content
Improved user experience	Better platform design
Targeted marketing	Useful advertisements
Community growth	Better interaction



Disadvantages



Disadvantages

Explanation

Privacy concerns

User data tracking

Addiction risk

Excessive social media use

Misuse of data

Personal data exploitation



Real-Life Social Computing Examples

Platform

Individual Behavior Example

Instagram

Reel watching and posting

YouTube

Video consumption and upload

Twitter/X

Tweet posting and retweeting

Facebook

Liking and sharing posts

LinkedIn

Professional content sharing



Memory Trick (Mnemonic)



"CPP"

C → Consumption

P → Participation



P → Production

👉 Three categories of online behavior.

✅ Exam Writing Tips

✅ Underline keywords:

- Individual Behavior
- Consumption
- Participation
- Production
- User Interaction

✅ Draw all three diagrams neatly.

✅ Mention Instagram and YouTube examples.

✅ Write comparison table.

✅ Use headings for each category.

👉 Diagrams + examples + comparison table = Extra marks.

✅ Short Conclusion

Individual online behavior represents how users interact with social media platforms. It mainly includes consumption, participation, and production behaviors, which help platforms analyze users and improve services.



Last Minute Revision Points

- Individual behavior = user online activity
- Three categories → CPP
- Consumption = viewing content
- Participation = likes/comments/shares
- Production = content creation
- Important in recommendation systems
- Used in Instagram and YouTube analysis
- Helps targeted marketing



Viva Questions with Answers

Q1. What is individual behavior?

Ans: Actions performed by users online.

Q2. Name three categories of online behavior.

Ans: Consumption, Participation, Production.

Q3. What is participation behavior?

Ans: Liking, commenting, and sharing content.



Q4. Give one example of production behavior.

Ans: Uploading YouTube videos.

Q5. Why is behavior analysis important?

Ans: To improve recommendations and user experience.

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What is Collective Behavior Analysis? Explain User Migration in social media.

9

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What is Collective Behavior Analysis? Explain User Migration in Social Media.

Simple Introduction

In social networks, users often behave in groups or communities.

Studying the behavior of many users together is called **Collective Behavior Analysis**.

👉 Simple meaning:

"Analyzing group behavior of users in social media."

Definition of Collective Behavior Analysis

Collective Behavior Analysis is the study of actions, interactions, trends, and activities performed by groups of users in online social networks.

It analyzes how communities behave together on platforms like:

- Instagram,
 - Facebook,
 - Twitter/X,
 - YouTube,
 - Reddit,
 - Telegram.
-

Real-Life Example

Twitter/X Trending Example

During a cricket match:

Millions of users tweet:

```
#IPLFinal  
#ViratKohli
```

👉 Large number of users showing similar behavior together is collective behavior.

YouTube Example

When a new movie trailer releases:

- millions watch it,
- share it,
- comment on it.

👉 Group activity forms collective behavior.

✅ Features of Collective Behavior Analysis

- Studies group activities
 - Detects trends
 - Analyzes communities
 - Finds viral content
 - Helps recommendation systems
-

✅ Simple Diagram of Collective Behavior

```
graph TD;
  Users --> Interactions;
  Interactions --> Group_Activities[Group Activities];
  Group_Activities --> Collective_Behavior_Analysis[Collective Behavior Analysis];
```

✅ Types of Collective Behavior in Social Media

Type	Example
Viral Trends	Trending hashtags
Online Communities	Gaming groups
Group Discussions	Reddit communities
Social Movements	Awareness campaigns



What is User Migration in Social Media?



Definition

User Migration means users moving from one social media platform or online community to another.



Simple meaning:

“Users shifting from one platform to another.”



Real-Life Examples of User Migration



Example 1 → WhatsApp to Telegram

Many users shifted from WhatsApp to Telegram because of:



- privacy concerns,
- better features,
- large group support.

👉 This movement is called **User Migration**.



Example 2 → Twitter/X to Threads

Some users moved from Twitter/X to Threads because of:

- new features,
 - better experience,
 - less spam.
-



Example 3 → Facebook to Instagram

Young users prefer Instagram because:

- reels,
- visual content,
- modern features.

👉 Users migrated from Facebook to Instagram.



Reasons for User Migration



Better Features

Users move to platforms with:

- better UI,
 - new features,
 - improved services.
-



Privacy and Security

Users migrate if they feel data is unsafe.



Popularity and Trends

People follow platforms where friends and influencers are active.



Less Spam and Advertisements

Too many ads may force users to switch platforms.

5 Faster Communication

Users prefer faster and smoother apps.

6 Community Influence

If a group/community shifts, many users follow them.

✓ Simple Diagram of User Migration

```
graph TD; A[Old Platform  
(WhatsApp)] --> B[User Shift]; B --> C[New Platform  
(Telegram)];
```

✓ Working of User Migration

```
graph TD; A[User Dissatisfaction] --> B[Search for Better Platform]; B --> C[Community Shift];
```




Migration to New Platform



Effects of User Migration

Positive Effects

Better user experience

Improved communication

Better privacy

More innovation

Negative Effects

Loss of users on old platform

Community fragmentation

Data transfer problems

Competition increase



Applications of Collective Behavior Analysis



Social Media Platforms

- Trend detection
- Viral content analysis
- Community analysis



Marketing

- Advertisement targeting



E-Commerce



- Customer behavior study



Politics

- Public opinion analysis



Education

- Student community analysis



Applications of User Migration Analysis

- Platform improvement
- Marketing strategy
- User retention
- Feature development
- Competitor analysis



Real-Life Social Computing Examples

Platform	Example
Instagram	Reels trend analysis
Twitter/X	Viral hashtag spread



Platform	Example
Telegram	Migration from WhatsApp
Reddit	Community movement
YouTube	Viewer community behavior

Advantages of Collective Behavior Analysis

Advantages	Explanation
Detects trends quickly	Finds viral topics
Improves recommendations	Better content suggestions
Helps business decisions	Understands audience
Supports marketing	Targeted advertising

Disadvantages

Disadvantages	Explanation
Privacy concerns	User data collection
Fake trends possible	Bot activities
Complex analysis	Huge data handling

Memory Trick (Mnemonic)

"TUG"

T → Trends

U → User groups

G → Group behavior

👉 Collective behavior studies groups and trends.

Memory Trick for User Migration

"SHIFT"

S → Switch platform

H → Higher features

I → Influence of community

F → Faster services

T → Trends

Exam Writing Tips

✓ Underline keywords:

- Collective Behavior
- User Migration
- Social Media Trends
- Community Analysis
- Platform Shift

✓ Draw migration diagram neatly.

✓ Mention WhatsApp → Telegram example.

✓ Use table format for effects.

✓ Write reasons point-wise.

👉 Examples + diagrams + tables = Extra marks.

✓ Short Conclusion

Collective Behavior Analysis studies how groups behave in social media, while User Migration explains movement of users between platforms. Both are important concepts in social computing and social network analysis.

✓ Last Minute Revision Points

- Collective behavior = group behavior analysis
- Helps detect trends and communities
- User migration = shifting between platforms

- Reasons include privacy and better features
 - WhatsApp → Telegram is common example
 - Used in marketing and recommendation systems
 - Important in social network analysis
 - Helps platform improvement
-

Viva Questions with Answers

Q1. What is collective behavior analysis?

Ans: Study of group behavior in social networks.

Q2. What is user migration?

Ans: Movement of users from one platform to another.

Q3. Give one example of user migration.

Ans: WhatsApp to Telegram.

Q4. Why do users migrate?

Ans: Better features and privacy.

Q5. What is trend analysis?

Ans: Finding popular topics and activities.

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Write short note on Scraping, Parsing and Crawling the web.

9

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Write Short Note on Scraping, Parsing and Crawling the Web



Simple Introduction

The internet contains huge amounts of data in the form of:

- web pages,
- social media posts,
- images,
- videos,
- comments,
- news articles.

To collect and analyze this data, techniques like:

- **Web Scraping**
- **Parsing**
- **Web Crawling**

are used.

👉 These methods help extract useful information from websites and social media.



1 Web Scraping



Definition

Web Scraping is the process of automatically extracting data from websites.

👉 Simple meaning:

“Collecting information from web pages automatically.”



Working of Web Scraping

```
Website
  ↓
Scraper Tool
  ↓
Extract Data
  ↓
Store Information
```



Real-Life Example



Twitter/X Example

A company collects tweets related to:

#IPL

#AI

using scraping tools.

👉 Tweets are extracted for trend analysis.



E-Commerce Example

Amazon product prices are collected automatically for price comparison.



Features of Web Scraping

- Automatic data collection
 - Fast extraction
 - Large-scale data gathering
 - Useful for analytics
-



Applications of Web Scraping

- Social media analysis
- Price comparison websites
- News collection
- Market research

- Trend analysis
-



2 Parsing



Definition

Parsing means analyzing and organizing extracted data into understandable format.

👉 Simple meaning:

“Breaking and understanding data properly.”



Working of Parsing

After scraping data:

Raw HTML Data



Parser



Structured Information



Real-Life Example



Instagram Example



Suppose scraped data contains:

```
<name>Ashish</name>  
<followers>1000</followers>
```

Parser extracts:

- username,
- followers count,
- post information.

👉 Converts raw data into useful format.

Features of Parsing

- Organizes raw data
- Converts HTML/XML data
- Makes data readable
- Helps data analysis

Applications of Parsing

- Search engines
- Social media analytics
- Web applications
- Recommendation systems



3 Web Crawling



Definition

Web Crawling is the process of automatically visiting and exploring web pages using bots called crawlers or spiders.

👉 Simple meaning:

“Automatically traveling through websites and collecting page links.”



Working of Web Crawling

```
Start URL
  ↓
Crawler Visits Page
  ↓
Find New Links
  ↓
Visit More Pages
```



Real-Life Example



Google Search Engine

Google crawler visits millions of web pages and indexes them for search results.



YouTube Example

YouTube crawls videos, titles, and descriptions to improve search recommendations.



Features of Web Crawling

- Automatically visits web pages
 - Discovers new pages
 - Creates search indexes
 - Collects linked information
-



Applications of Web Crawling

- Search engines
 - Website indexing
 - SEO analysis
 - Social media monitoring
 - Trend discovery
-



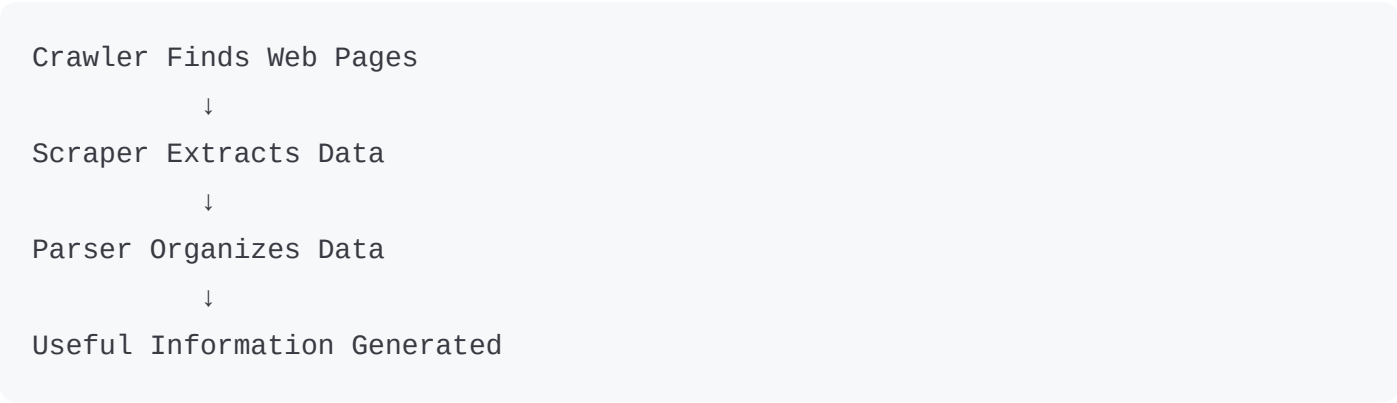
Difference Between Scraping, Parsing and Crawling



Scraping	Parsing	Crawling
Extracts data	Organizes data	Discovers web pages
Collects information	Analyzes structure	Navigates websites
Example: Tweet collection	Example: HTML analysis	Example: Google bot



Combined Working Process



Real-Life Social Computing Examples

Platform	Use
Twitter/X	Tweet scraping
Instagram	User profile analysis
YouTube	Video indexing
Facebook	Community data analysis
LinkedIn	Job data extraction

Advantages

Advantages	Explanation
Fast data collection	Saves time
Large-scale analysis	Handles huge data
Useful for research	Helps data mining
Better recommendations	Improves services

Disadvantages

Disadvantages	Explanation
Privacy concerns	User data misuse
Legal issues	Unauthorized scraping
High server load	Excess crawling affects websites
Complex implementation	Difficult for beginners

Memory Trick (Mnemonic)

 "CSP"

C → **C**rawling finds pages

S → Scraping extracts data

P → Parsing organizes data

👉 Order of web data processing.

Exam Writing Tips

 Underline keywords:

- Web Scraping
- Parsing
- Crawling
- Crawler
- Parser

 Draw combined process diagram.

 Mention Google crawler example.

 Write comparison table neatly.

 Use Instagram/Twitter examples.

👉 Diagram + comparison table + examples = Extra marks.

Short Conclusion

Scraping, parsing, and crawling are important web data collection techniques used in search engines, social media analysis, recommendation systems, and web mining applications.



Last Minute Revision Points

- Crawling discovers web pages
 - Scraping extracts web data
 - Parsing organizes raw data
 - Google uses crawlers
 - Social media uses scraping
 - Parsing converts HTML into readable data
 - Used in web mining and analytics
 - Important for search engines
-



Viva Questions with Answers

Q1. What is web scraping?

Ans: Automatic extraction of website data.

Q2. What is parsing?

Ans: Organizing raw data into useful format.

Q3. What is web crawling?

Ans: Automatic visiting of web pages.



Q4. Which search engine uses crawlers?

Ans: Google.

Q5. What is a crawler also called?

Ans: Spider or bot.

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Describe breadth-first search in web crawling and pros & cons.
9

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Describe Breadth-First Search in Web Crawling and Its Pros & Cons

Simple Introduction

Web crawling is used to automatically visit and collect information from websites.

To decide which web pages should be visited first, crawling algorithms are used.

One important crawling method is:

Breadth-First Search (BFS)

👉 BFS visits nearby web pages level-by-level.

Definition of Breadth-First Search (BFS)

Breadth-First Search (BFS) is a web crawling technique in which the crawler first visits all neighboring web pages before moving to the next level of pages.

👉 Simple meaning:

"Visit all nearby pages first, then go deeper."

✅ Real-Life Example



Google Search Engine Example

Suppose crawler starts from:

www.news.com

It first visits all links present on homepage.

Then visits links inside those pages.

👉 This is BFS crawling.

✅ Working of BFS in Web Crawling



Step-by-Step Process

Step 1 Start from Initial Web Page

Crawler starts with one URL.

Example:

Page A

Step 2 Visit All Neighbor Pages

Crawler checks all links connected to Page A.

Example:

A → B, C, D

Step 3 Store Links in Queue

All discovered pages are stored in queue.

Step 4 Visit Pages One-by-One

Crawler visits:

B → C → D

before going deeper.

Step 5 Continue Level-by-Level

After finishing first level:

Crawler moves to next level pages.

✓ Simple Diagram of BFS Crawling



✓ BFS Visiting Order

A → B → C → D → E → F

👉 Same-level pages visited first.

✓ Queue Structure in BFS

BFS uses:

📌 Queue (FIFO)

FIFO means:

First In → First Out

👉 First discovered page is visited first.

✅ BFS Algorithm for Web Crawling

📌 Simple Steps

1. Start with root URL
2. Add URL to queue
3. Visit page
4. Extract links
5. Add new links to queue
6. Repeat process

✅ Example of BFS Crawling

📱 YouTube Example

Crawler starts from:

YouTube Homepage

First visits:

- Trending page
- Shorts page
- Music page

Then explores links inside them.

👉 Nearby pages are visited first.

✅ Applications of BFS in Web Crawling

🌐 Search Engines

- Google
- Bing



Social Media Platforms

- Instagram content crawling
- Twitter trend collection



E-Commerce

- Product page indexing



News Websites

- Article collection
-

✅ Advantages of BFS Crawling

Advantages	Explanation
Finds nearby pages quickly	Important pages discovered early
Simple implementation	Easy algorithm
Good for shallow web pages	Efficient for close pages
Better page coverage	Visits all same-level pages

Disadvantages of BFS Crawling

Disadvantages	Explanation
High memory usage	Queue becomes large
Slow for deep pages	Deep pages visited late
May crawl unnecessary pages	Time wastage
Not priority-based	Important pages may wait

Difference Between BFS and DFS Crawling

BFS	DFS
Level-by-level traversal	Depth-wise traversal
Uses Queue	Uses Stack
Nearby pages first	Deep pages first
More memory needed	Less memory needed

Features of BFS Crawling

- Level-wise crawling
 - Uses queue data structure
 - Systematic page visiting
 - Good web coverage
 - Common in search engines
-

Real-Life Social Computing Examples

Platform	BFS Use
Google	Search indexing
YouTube	Video link crawling
Twitter/X	Trend page crawling
Instagram	Hashtag crawling
News Websites	Article indexing

Memory Trick (Mnemonic)

 **"LEVEL"**

L → Level-wise visiting

E → Explore neighbors first

V → Visit nearby pages

E → Expand queue

L → Link traversal

👉 BFS works level-by-level.

Exam Writing Tips

 Underline keywords:

- Breadth-First Search
- Web Crawling
- Queue
- FIFO
- Level-wise traversal

 Draw BFS tree diagram neatly.

 Mention Google crawler example.

 Write BFS visiting order.

 Use table format for pros & cons.

👉 Diagram + algorithm + comparison = Extra marks.

Short Conclusion

Breadth-First Search is an important web crawling technique where web pages are visited level-by-level using a queue. It is widely used in search engines and social media data collection systems.

Last Minute Revision Points

- BFS = level-wise crawling
 - Uses Queue (FIFO)
 - Visits nearby pages first
 - Common in Google crawling
 - Good page coverage
 - High memory usage
 - Better for shallow pages
 - Used in search engines
-

Viva Questions with Answers

Q1. What is BFS in web crawling?

Ans: Level-wise visiting of web pages.

Q2. Which data structure is used in BFS?

Ans: Queue.

Q3. What does FIFO mean?

Ans: First In First Out.

Q4. Which search engine uses crawling?

Ans: Google.

Q5. What is the main advantage of BFS?

Ans: Quickly visits nearby pages.

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Explain query human language data with TF-IDF.
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Explain Query Human Language Data with TF-IDF



Simple Introduction

Humans search information using normal language words like:

- "Java tutorial"
- "Best gaming laptop"
- "AI notes"

Search engines and social media platforms must understand these words and find the most relevant data.

To measure the importance of words in documents, a technique called **TF-IDF** is used.

👉 Simple meaning:

"TF-IDF helps identify important words in text data."



Definition of TF-IDF



TF-IDF (Term Frequency - Inverse Document Frequency) is a text mining technique used to measure how important a word is in a document compared to all other documents.

Meaning of Query Human Language Data

A **query** is the text entered by user in natural human language.

Example:

```
"Best Python tutorial"
```

The system analyzes these words and finds matching documents or web pages.

Real-Life Example



YouTube Search Example

Suppose user searches:

```
"Java DSA tutorial"
```

YouTube checks:

- titles,
- descriptions,
- tags,

- comments.

TF-IDF identifies important words like:

```
Java  
DSA  
Tutorial
```

👉 Videos containing these important words rank higher.



Google Search Example

When searching:

```
"IPL live score"
```

Google gives pages where these words are highly important.



Components of TF-IDF

TF-IDF has two parts:



TF → Term Frequency



IDF → Inverse Document Frequency





1 Term Frequency (TF)



Meaning

TF measures how many times a word appears in a document.



Formula of TF

$$TF = \frac{\text{Number of times term appears in document}}{\text{Total number of words in document}}$$



Example of TF

Document:

"Java is easy. Java is powerful."

Word:

Java

appears 2 times.

👉 Higher frequency = more importance in that document.



**2**

Inverse Document Frequency (IDF)



Meaning

IDF checks whether word is common or rare across all documents.

Common words like:

is, the, and

have low importance.

Rare words have high importance.



Formula of IDF

$$IDF = \log \left(\frac{\text{Total number of documents}}{\text{Number of documents containing the term}} \right)$$



Example of IDF

Word:

Java

appears in fewer documents.



👉 Higher IDF value.

Word:

the

appears everywhere.

👉 Lower IDF value.



Final TF-IDF Formula

$$TF-IDF = TF \times IDF$$

👉 Important words get higher TF-IDF score.



Working of TF-IDF



Step-by-Step Process

User Query



Text Analysis



Calculate TF



Calculate IDF



Compute TF-IDF Score





Rank Relevant Documents

✓ Simple Diagram

Query Words
↓
TF-IDF Calculation
↓
Important Keywords
↓
Relevant Results

✓ Example of TF-IDF Calculation

Suppose:

Document	Text
D1	Java tutorial for beginners
D2	Python programming tutorial
D3	Java DSA tutorial

User query:

Java tutorial

Word "Java" appears in fewer documents.

👉 Higher TF-IDF score.

So D1 and D3 rank higher.

Applications of TF-IDF

Search Engines

- Google search ranking

Social Media

- Hashtag analysis
- Tweet analysis

YouTube

- Video recommendation

Email Systems

- Spam detection

News Websites

- Article recommendation
-

Real-Life Social Computing Examples

Platform	TF-IDF Use
Google	Search ranking
YouTube	Video search
Twitter/X	Trend keyword analysis
Instagram	Hashtag importance
Reddit	Topic analysis

Features of TF-IDF

- Identifies important words
- Removes common meaningless words
- Improves search accuracy
- Supports text mining
- Helps answer ranking

Advantages of TF-IDF

Advantages	Explanation
Easy implementation	Simple mathematical method
Better search accuracy	Relevant results shown

Advantages	Explanation
Finds important keywords	Useful in text analysis
Reduces noise words	Ignores common words

✅ Disadvantages of TF-IDF

Disadvantages	Explanation
Ignores word meaning	Only frequency considered
Cannot understand context	Semantic meaning missing
Large datasets expensive	High computation

✅ Memory Trick (Mnemonic)

📌 "FIT"

F → **Frequency**

I → **Importance**

T → **Term analysis**

👉 TF-IDF measures important terms.

Exam Writing Tips

 Underline keywords:

- TF-IDF
- Term Frequency
- Inverse Document Frequency
- Query
- Ranking

 Write all formulas clearly.

 Draw TF-IDF process diagram.

 Mention Google and YouTube examples.

 Explain TF and IDF separately.

 Formula + example + diagram = Extra marks.

Short Conclusion

TF-IDF is an important text mining technique used to identify important words in documents. It improves search accuracy, query processing, and ranking in search engines and social media platforms.

Last Minute Revision Points

- TF-IDF measures keyword importance

- TF = word frequency in document
 - IDF = rarity of word in all documents
 - $TF-IDF = TF \times IDF$
 - Used in search engines
 - Improves answer ranking
 - Common in Google and YouTube
 - Important in text mining
-

Viva Questions with Answers

Q1. What is TF-IDF?

Ans: Technique to measure importance of words.

Q2. What does TF mean?

Ans: Term Frequency.

Q3. What does IDF mean?

Ans: Inverse Document Frequency.

Q4. Where is TF-IDF used?

Ans: Search engines and text mining.

Q5. What is query data?

Ans: User-entered search text.

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